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Quintero

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(54) **METHODS AND DEVICES FOR SKIN CLOSURE**

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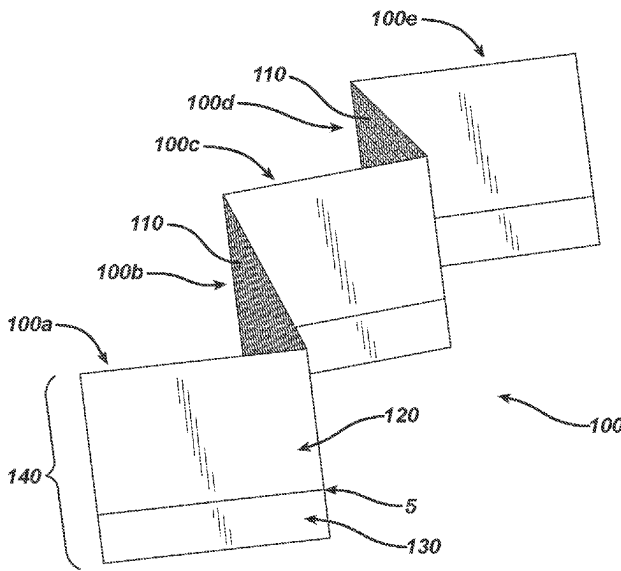
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(57) **ABSTRACT**

 A wound closure device and a method of its use are disclosed. The wound closure device comprises a wound closure strip, the wound closure strip comprising a wound-facing side and a top side, the wound-facing side comprises an adhesive applied over at least a portion of the wound facing side of the wound closure strip and a non-symmetric, two-part release liner assembly detachably adhered to the adhesive, the release liner assembly comprising a first section and a second section with the second section comprising a wound closure strip-free portion forming a tab. In an alternate embodiment, the device is multi-segmented and the wound closure strip may be coextensive with the release liner assembly.

9 Claims, 18 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/675,159, filed on Aug. 11, 2017, now Pat. No. 10,687,986, which is a continuation-in-part of application No. 15/280,303, filed on Sep. 29, 2016, now Pat. No. 10,470,934.

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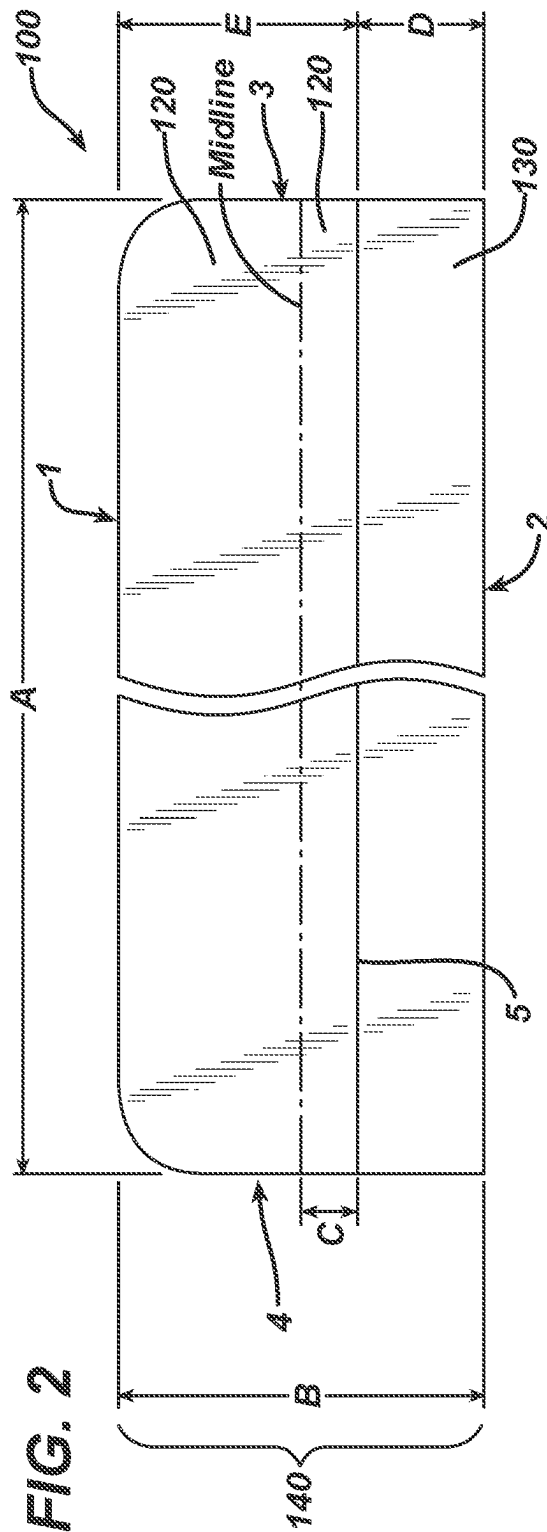
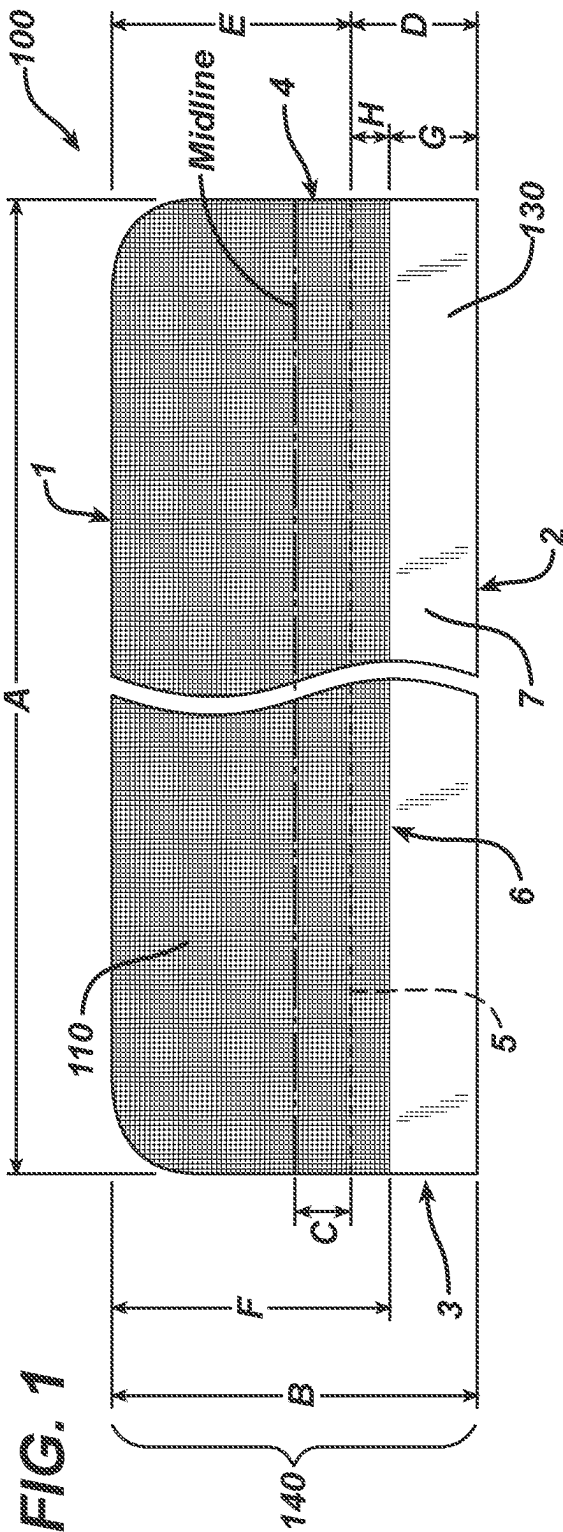
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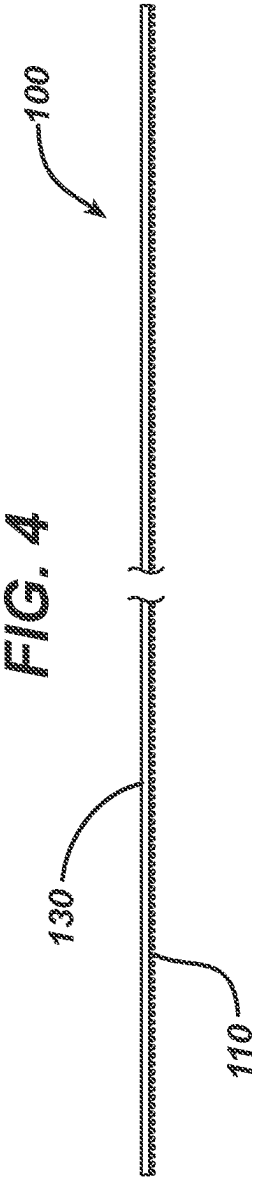
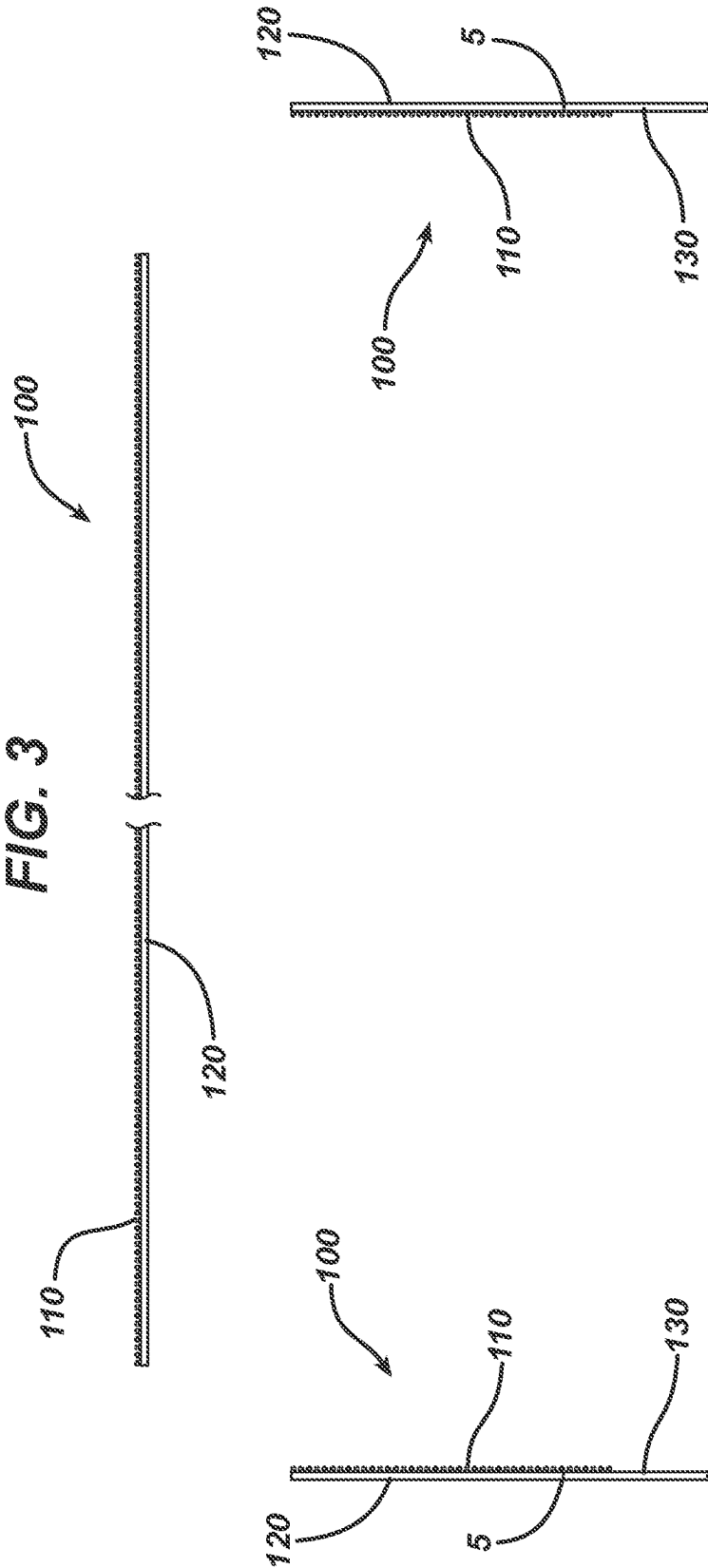
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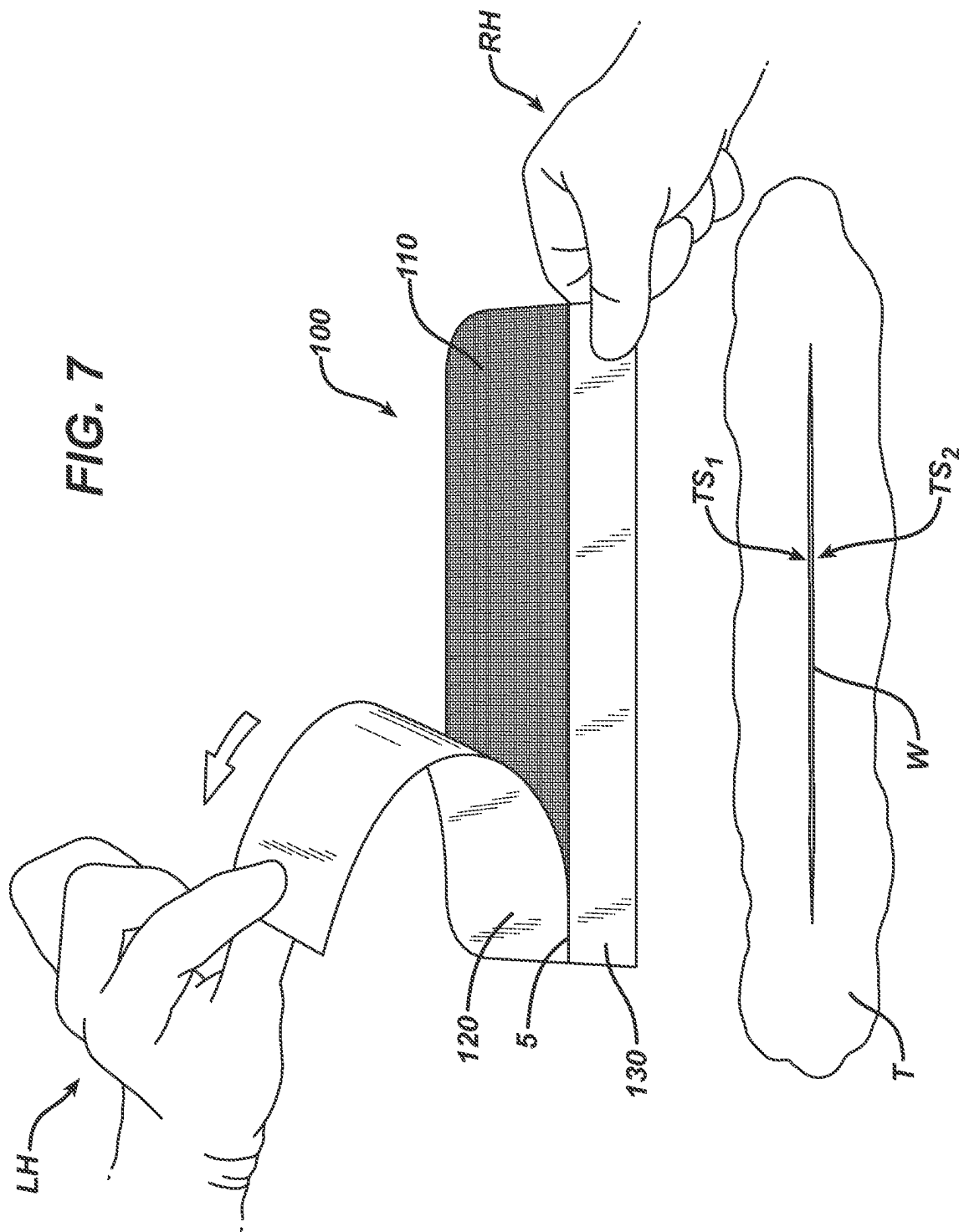


FIG. 8

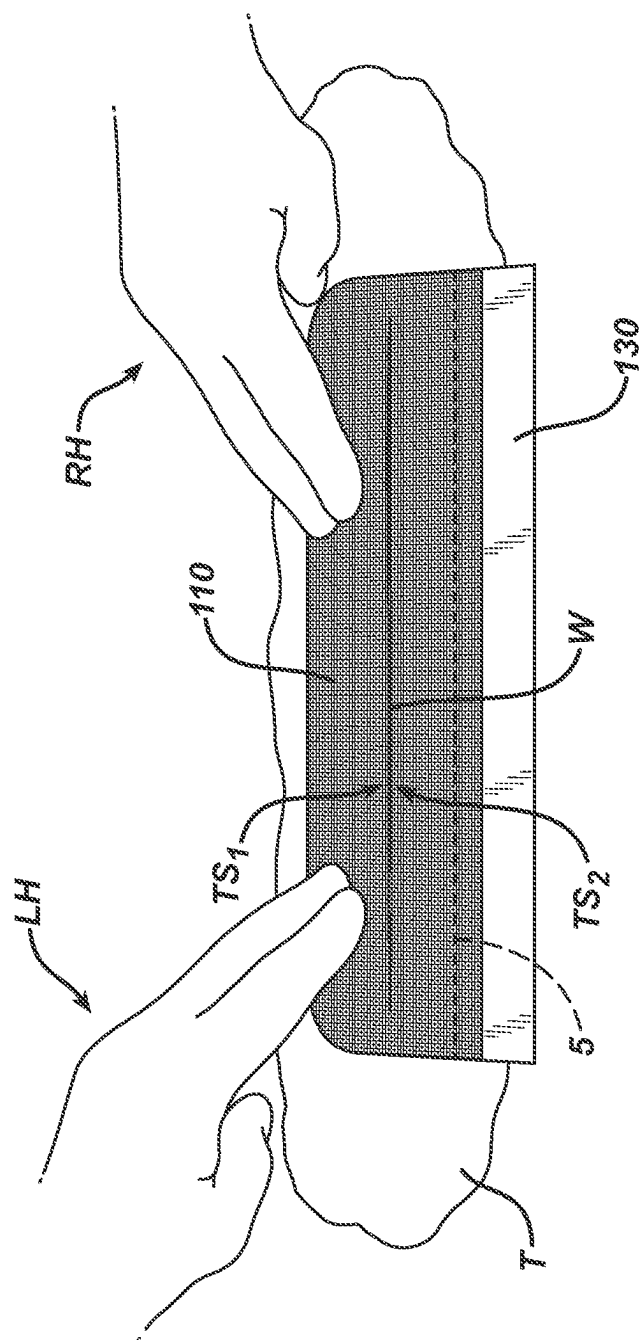


FIG. 9

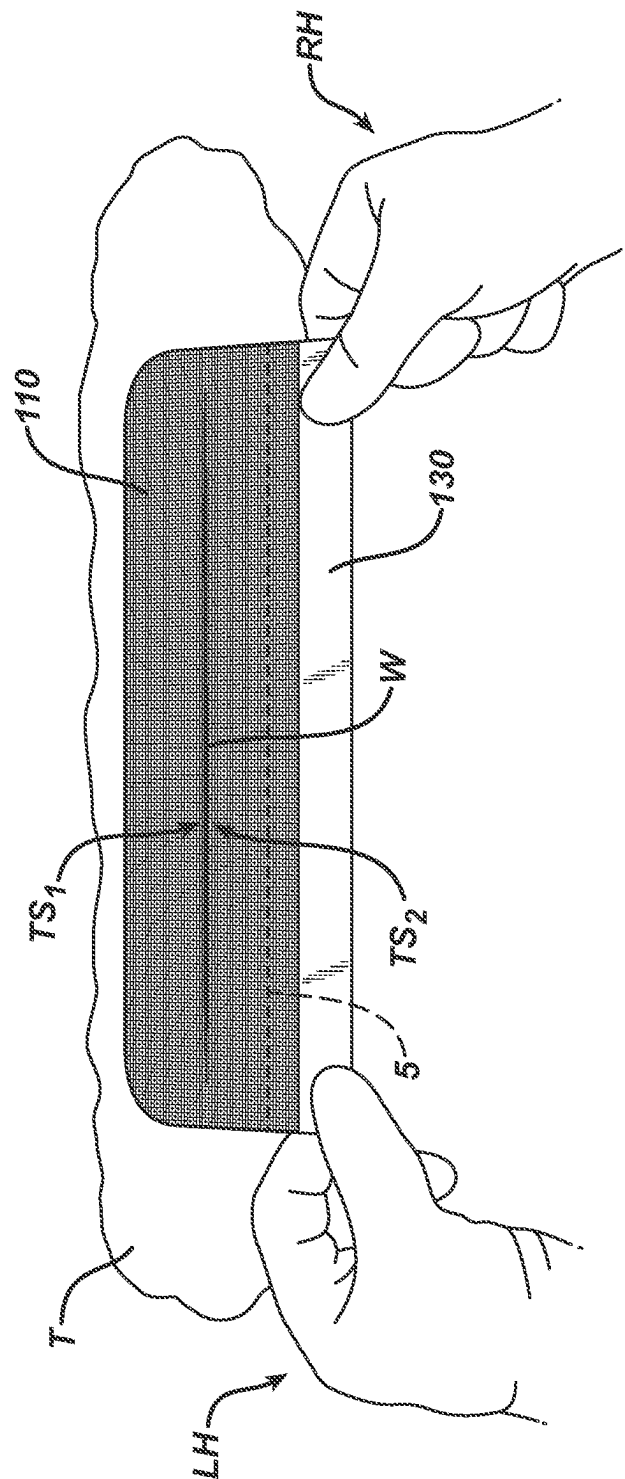


FIG. 10

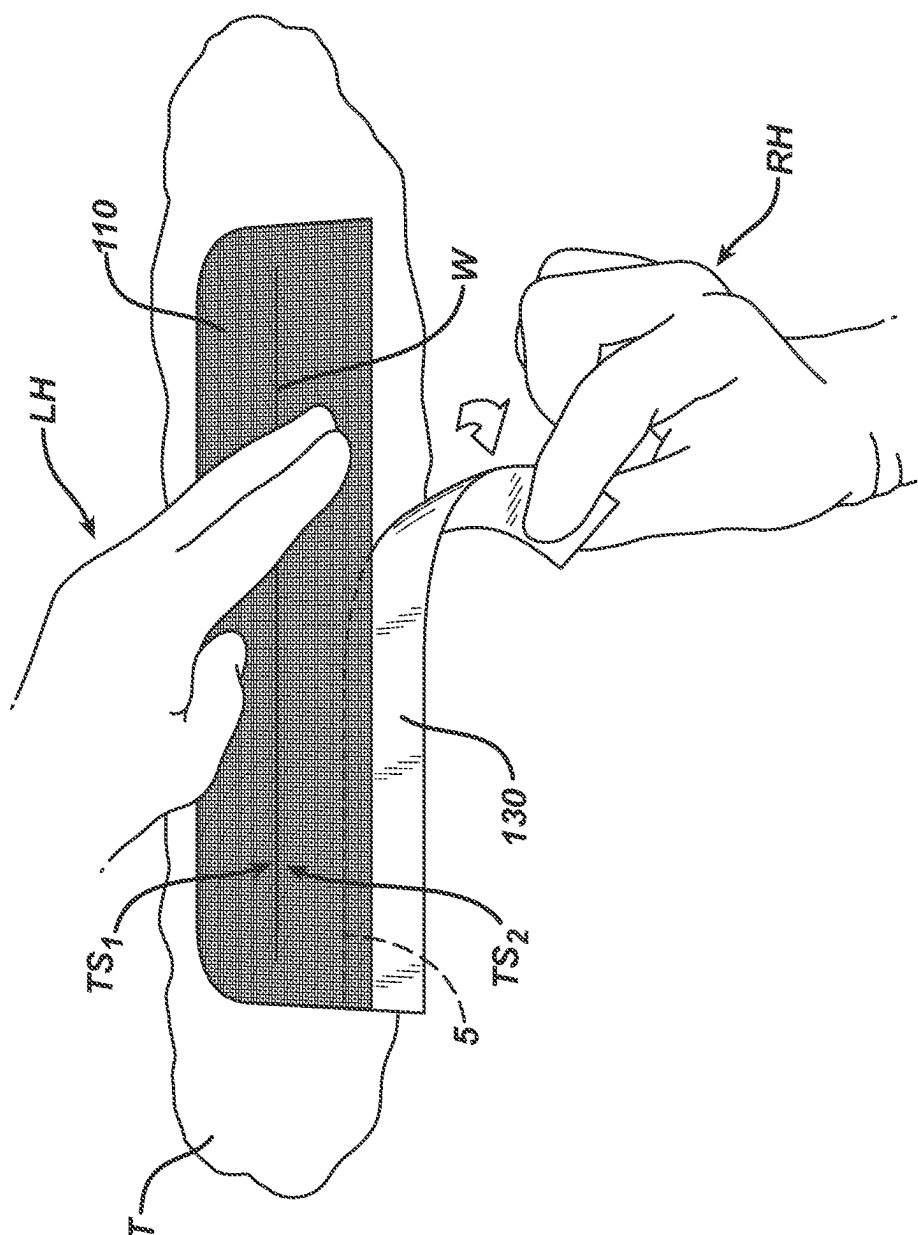


FIG. 11

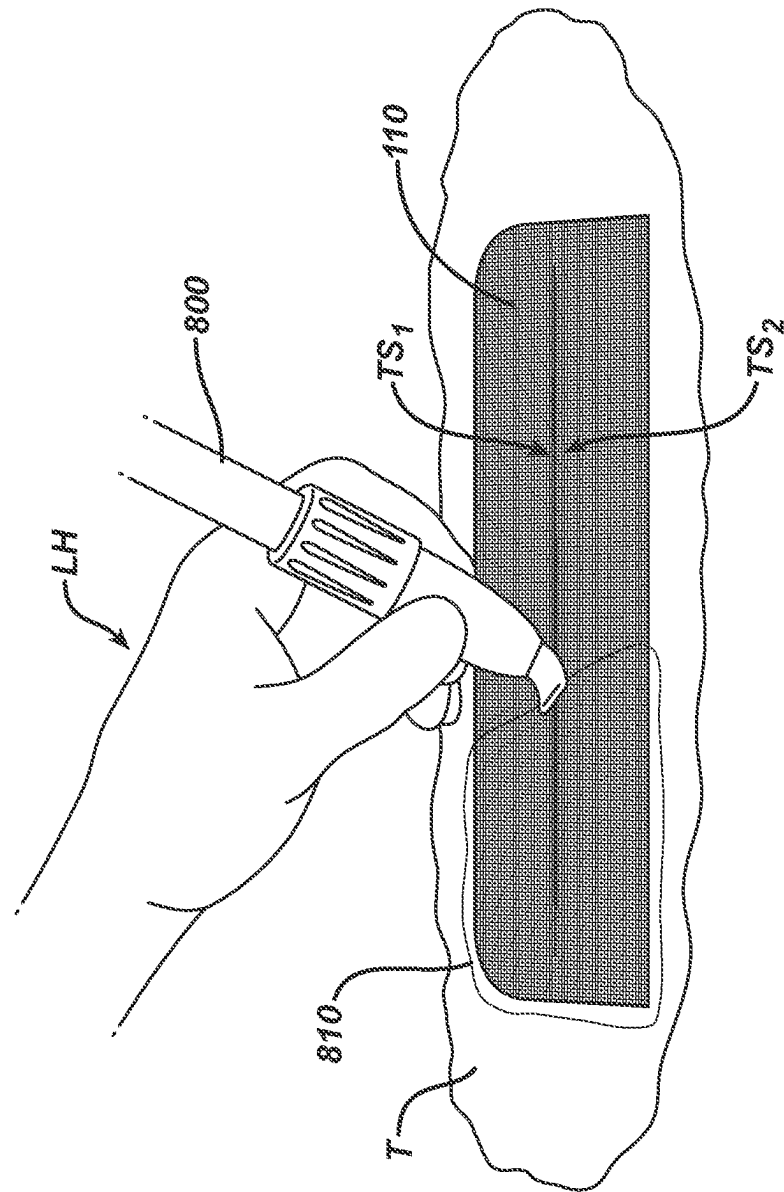


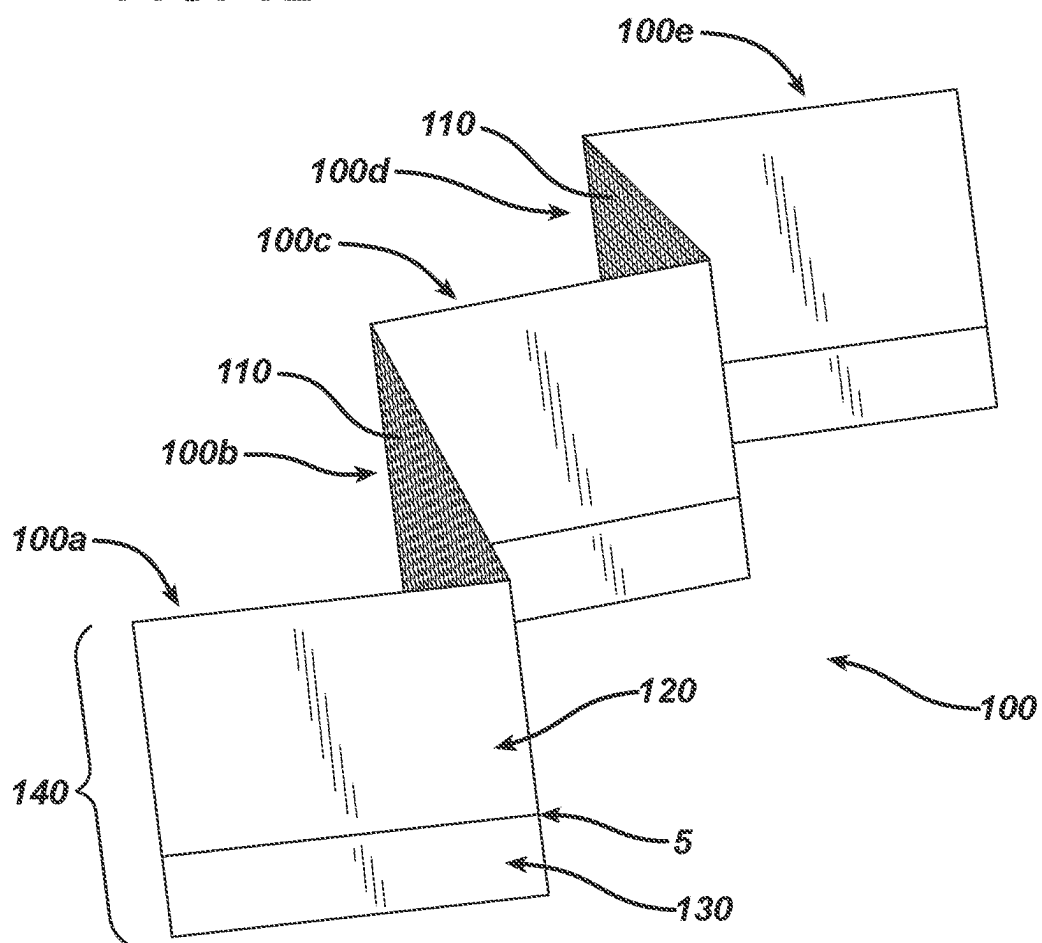
FIG. 12

FIG. 13

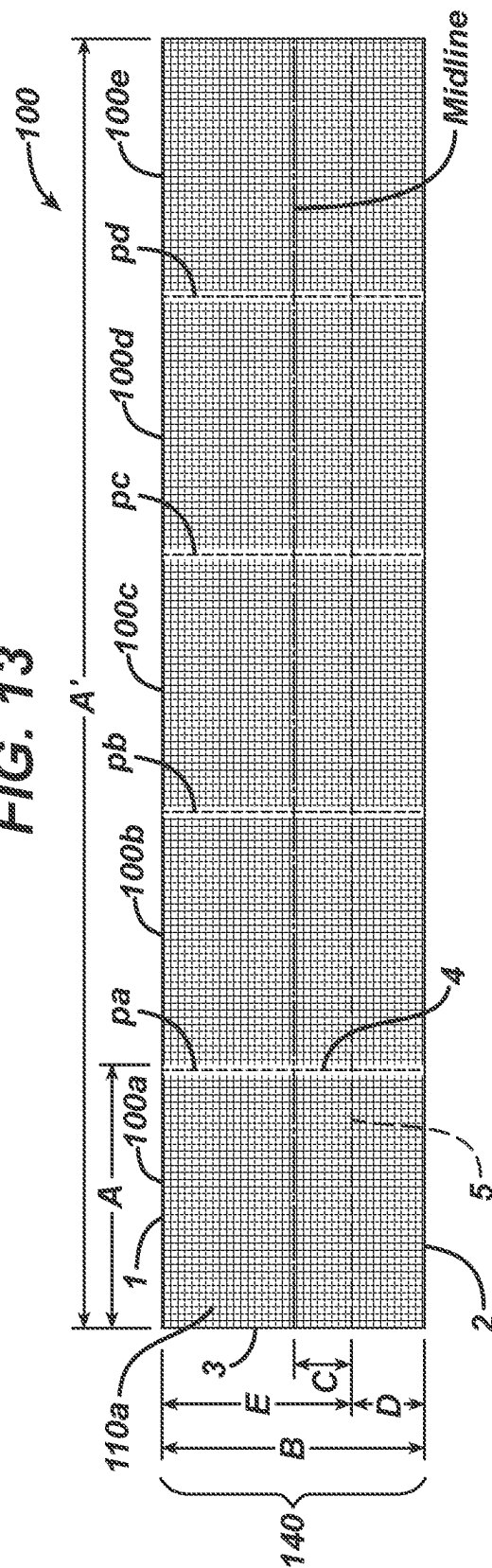


FIG. 14

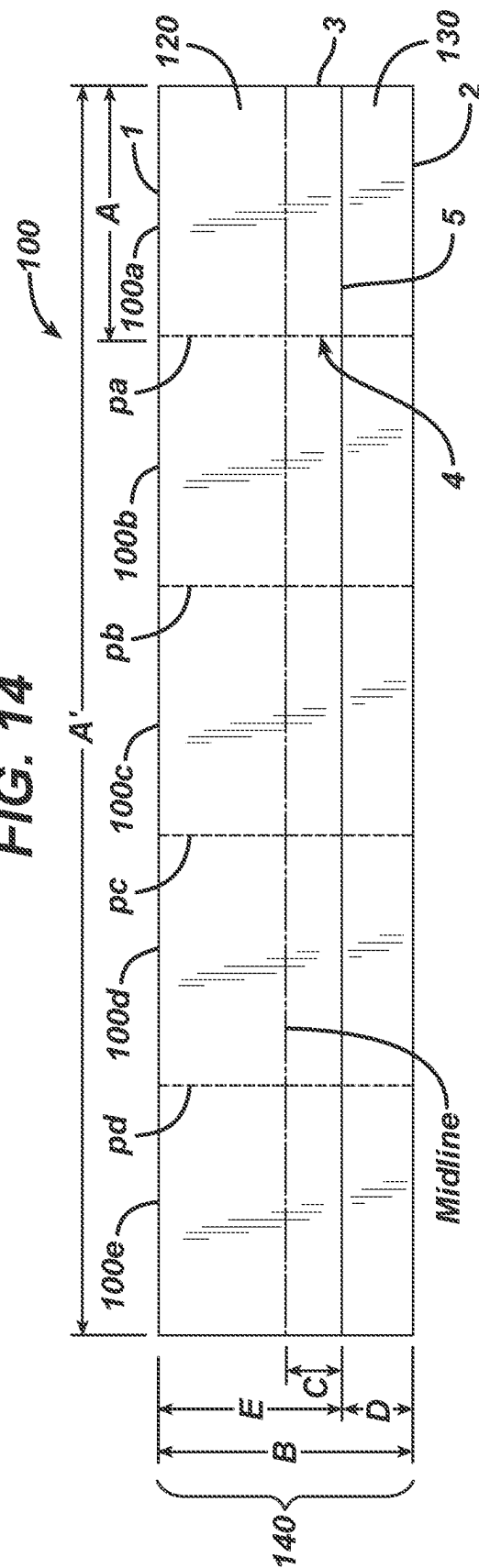


FIG. 15

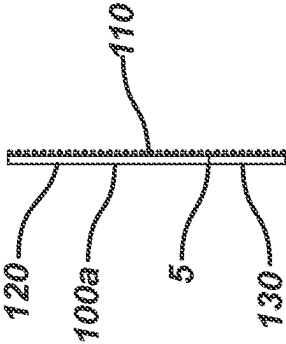
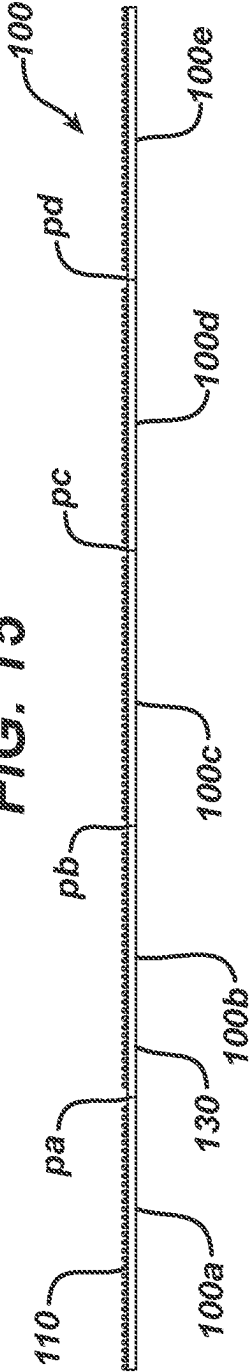


FIG. 17

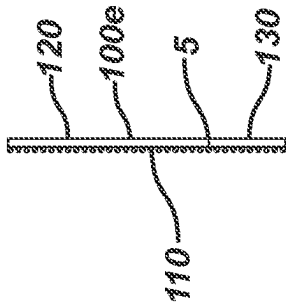


FIG. 18

FIG. 16

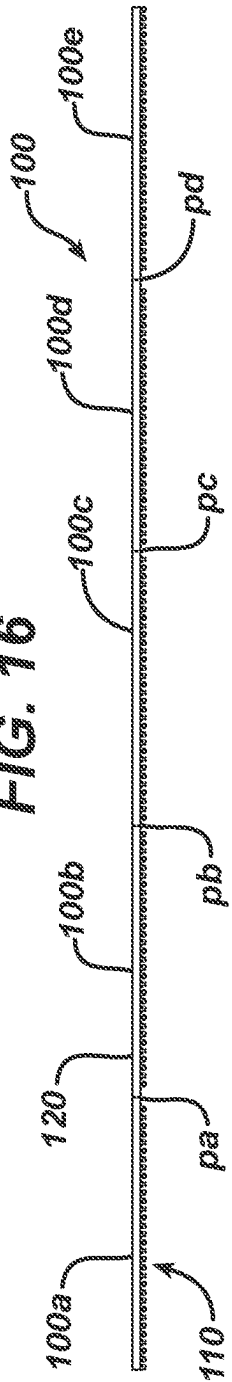


FIG. 19

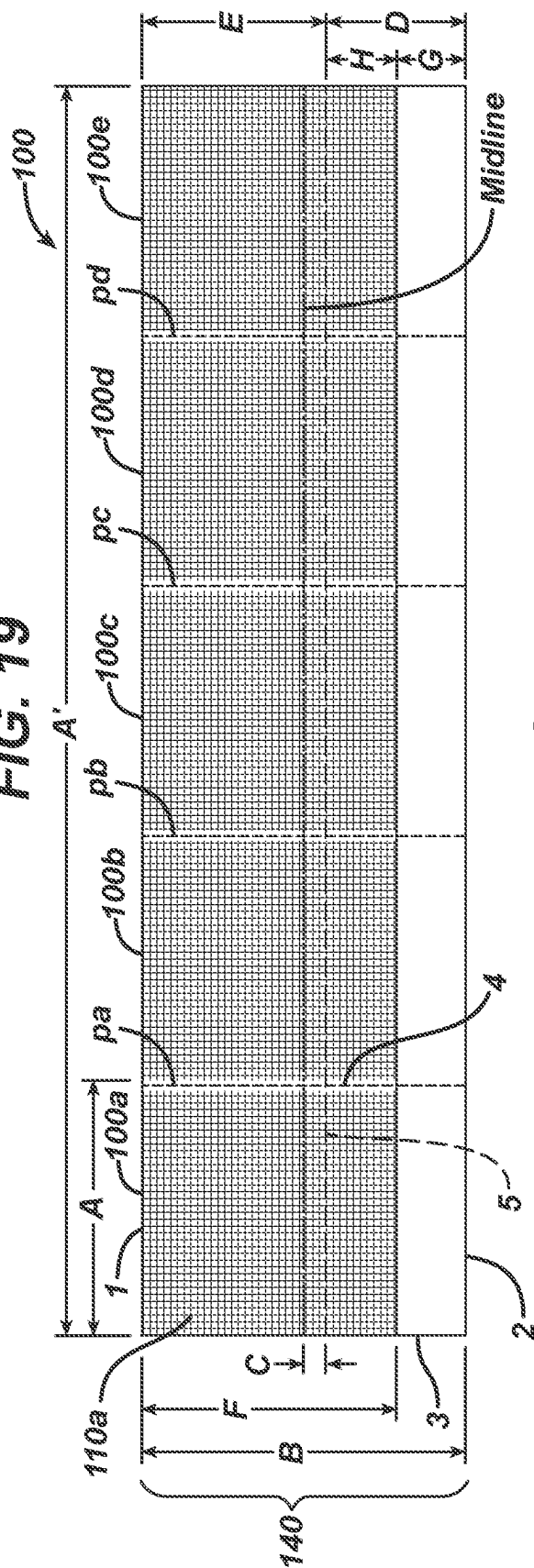


FIG. 20

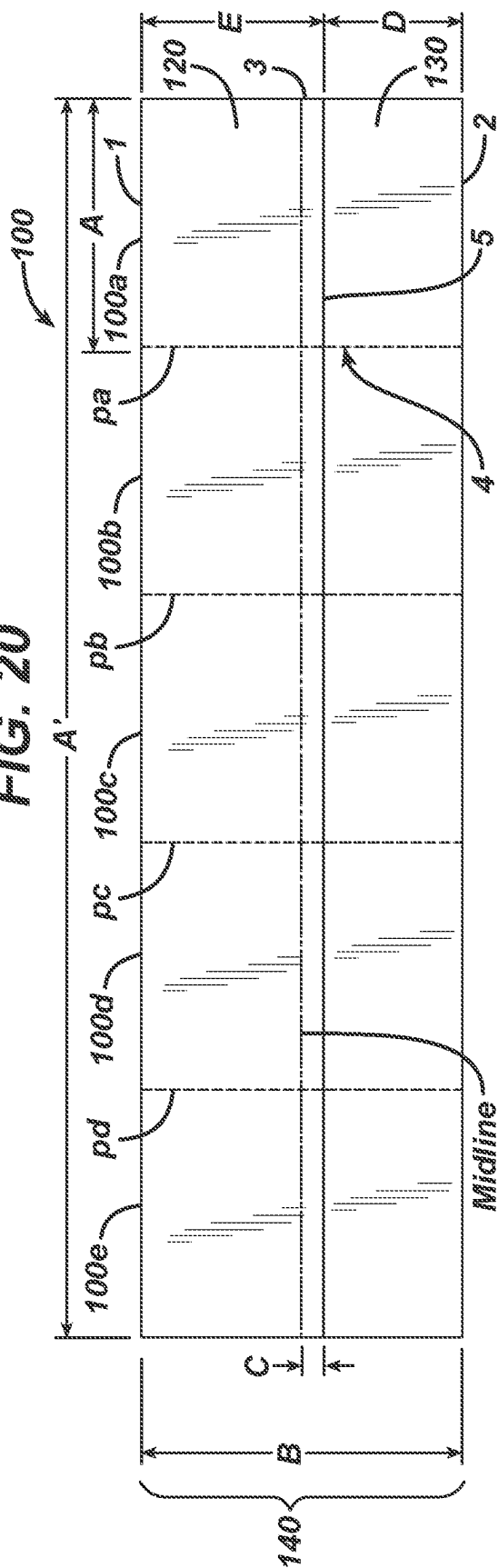


FIG. 21

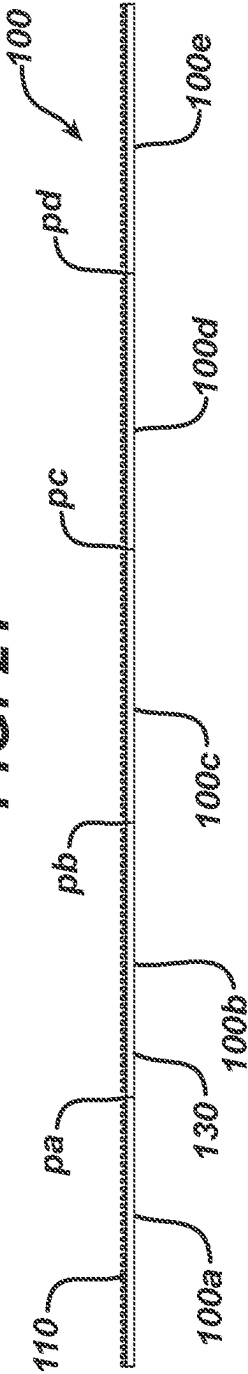


FIG. 23

FIG. 22

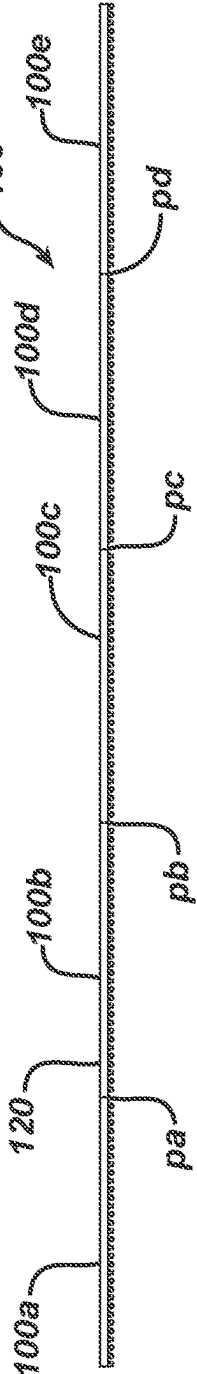


FIG. 24

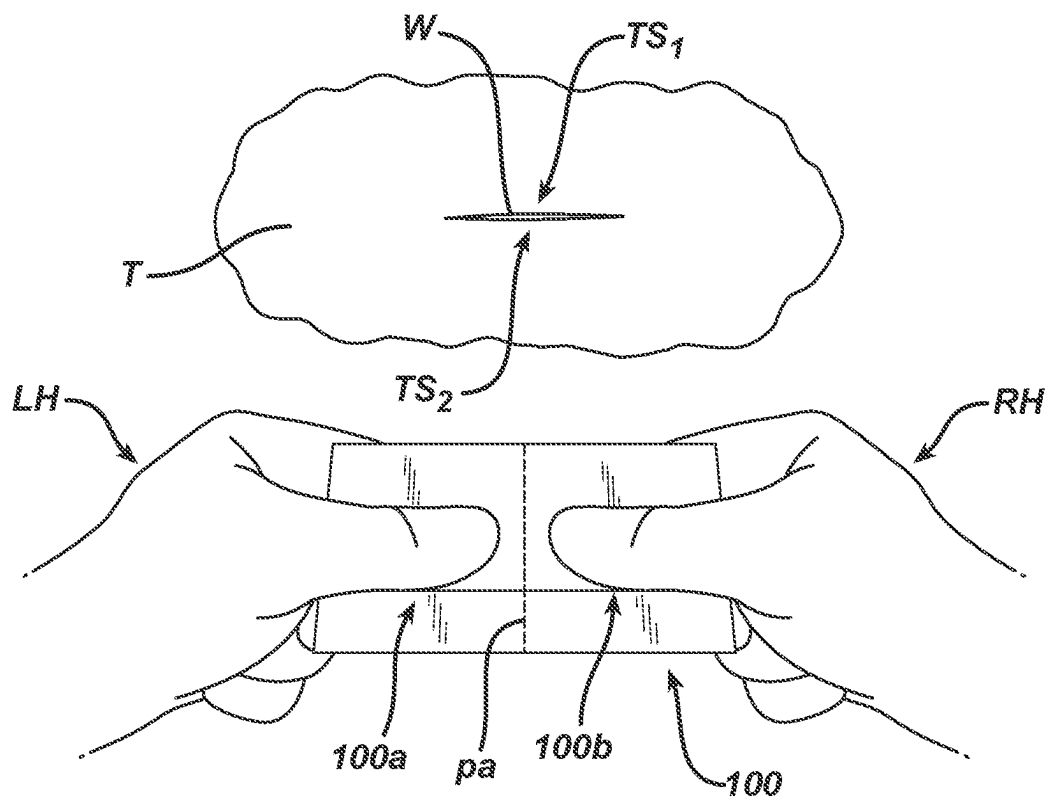
FIG. 25

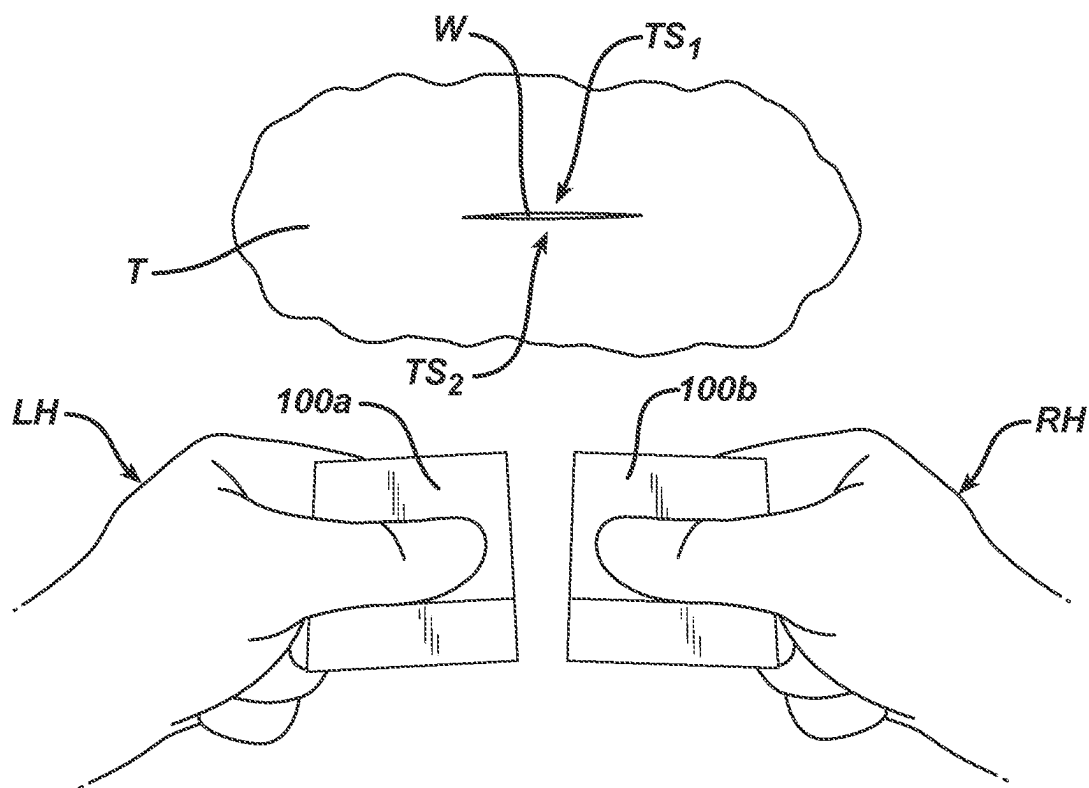
FIG. 26

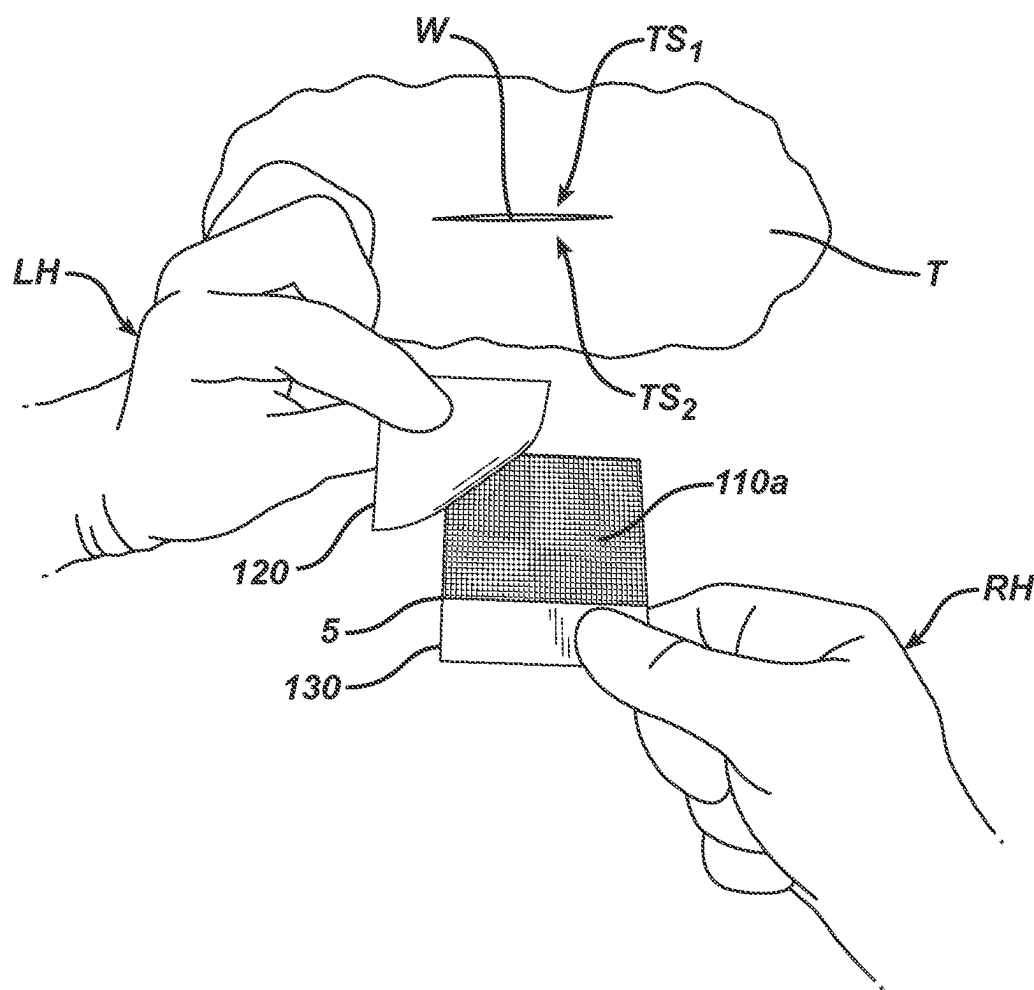
FIG. 27

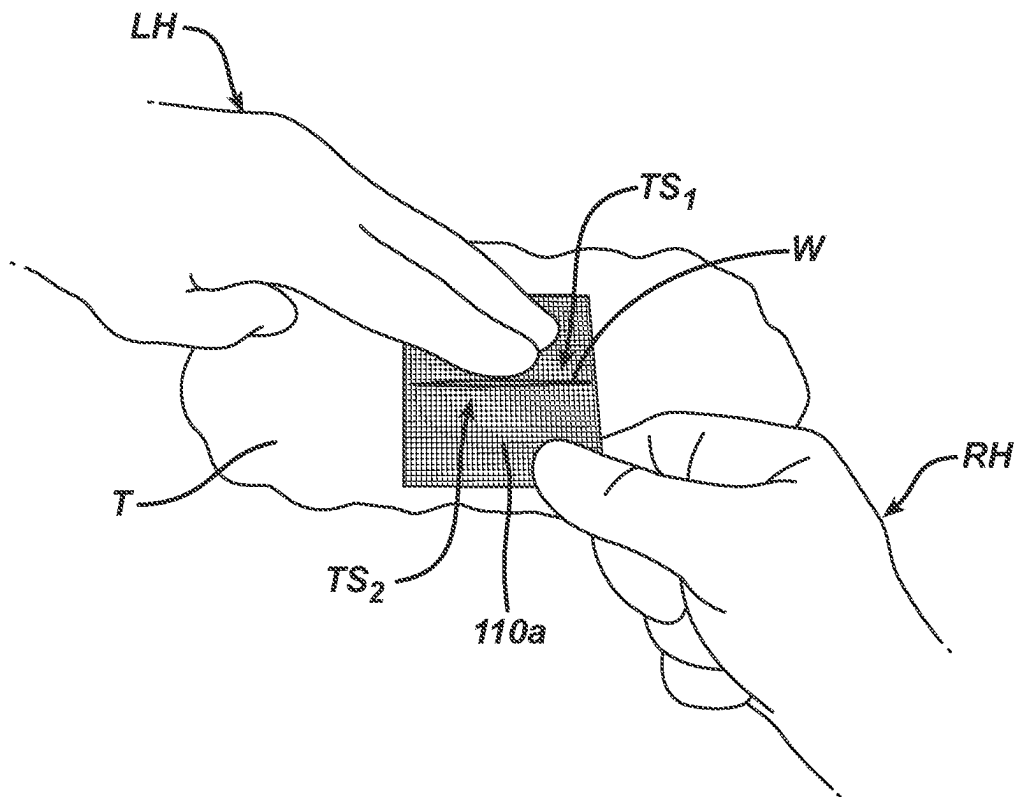
FIG. 28

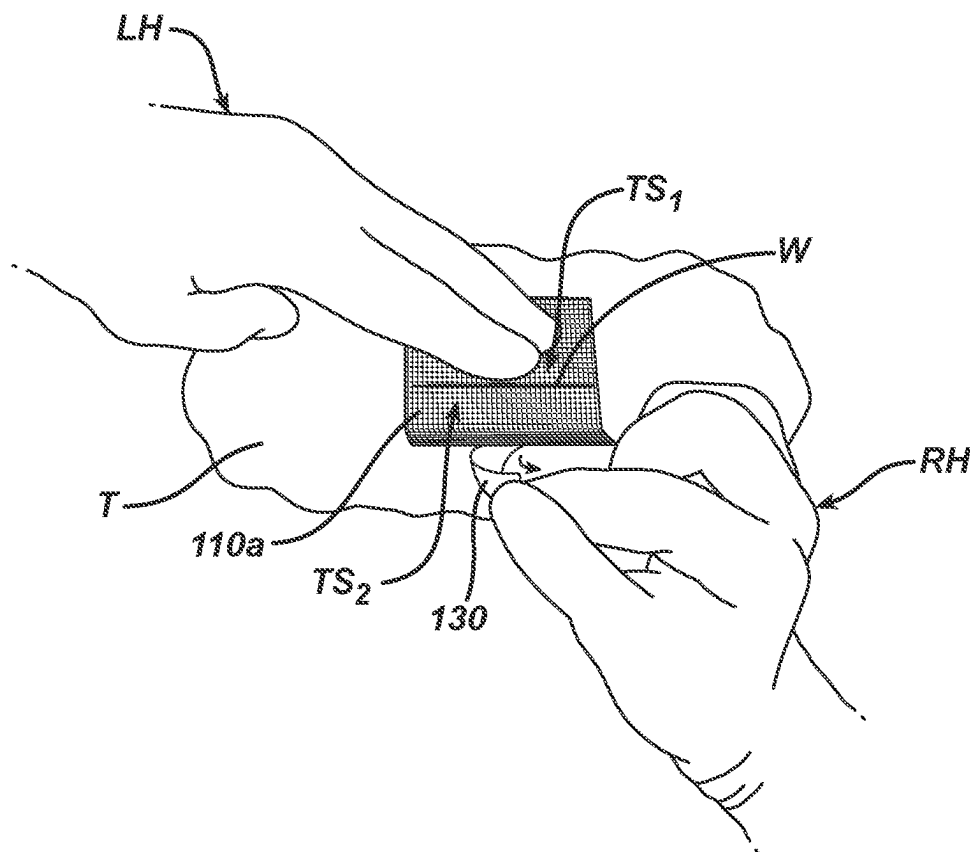
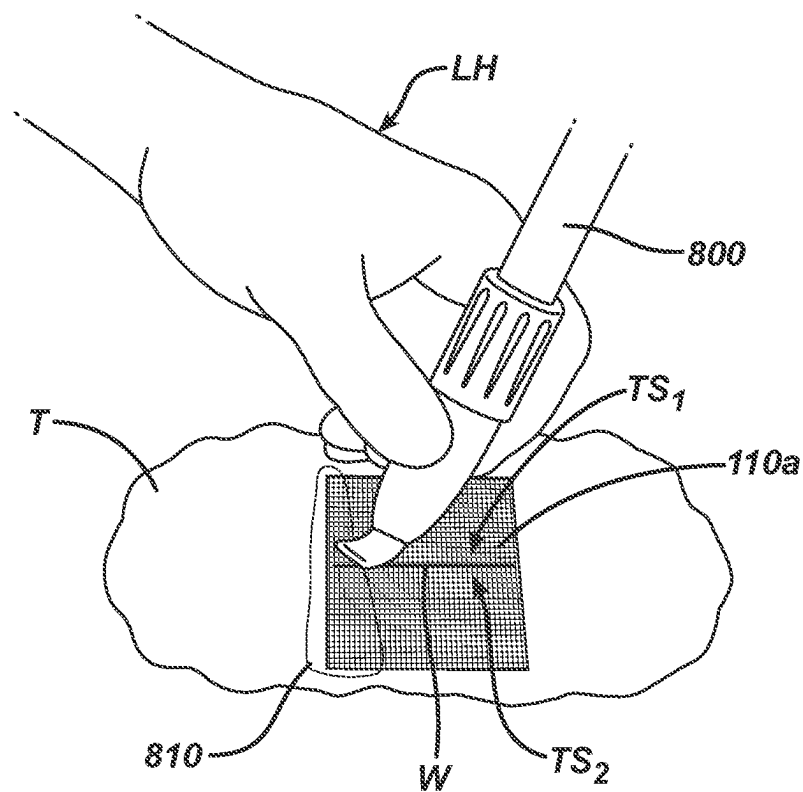
FIG. 29

FIG. 30

METHODS AND DEVICES FOR SKIN CLOSURE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of and claims the benefit of U.S. patent application Ser. No. 16/907,930, filed Jun. 22, 2020, which is a continuation of and claims the benefit of U.S. patent application Ser. No. 15/675,159, filed Aug. 11, 2027, now U.S. Pat. No. 10,687,986, which is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 15/280,303, filed Sep. 29, 2016, now U.S. Pat. No. 10,470,934. The complete disclosures of the aforementioned related U.S. patent application is hereby incorporated herein by reference for all purposes.

BACKGROUND OF THE INVENTION

Field of the Invention

This invention is concerned with devices, systems and methods for wound closure, particularly for skin closure of wounds formed by lacerations or surgical incisions.

Related Art

Wound closure tapes and topical adhesives provide an alternative to wound closure by staples and sutures. Among the advantages of using wound closure tapes or topical adhesives, and their combinations include less tissue trauma, improved cosmetic outcomes, and less pain compared with staples and sutures when treating skin closures.

US 2002/0193721 discloses a wound closure grip-like tape apparatus and methods of its use. Reference to FIG. 10 of US 2002/0193721 and the corresponding text of this publication shows how the grip tape is used to secure the wound of FIG. 9. In particular, paragraph 43 of the publication describes three techniques of wound closure reproduced in part as follows: "The first exemplary means of sealing the wound is to place WCGT ("Wound Closure Grid Tape") 100 on one side of the wound 90 with a slight bend along the axis between numbers 92 and 93 such that the other side 91 is tilted above the skin and thus will not adhere. With side 90 pressed in place and with the appropriate pressure applied, the WCGT is tugged with one hand while the other hand (or the thumb of the hand if being self-applied or with one hand) pushes the skin on the other side 91 into position, and then the WCGT is lowered into place. This handling can be done by grasping the WCGT at the edges where denigration of the adhesive is not as critical, or alternately by leaving the backing attached to the secondary side 91 while the first side is pressed into place 90. The second exemplary means to apply is to curve the WCGT upwards, or hold it in a curved position upwards along the longer axis shown, and position it starting at one end of the wound 92. Then, as previously, the wound is pressed together and the WCGT is pressed into place gradually along the length of the wound, going from one end 92 to the other 93. For small wounds, a third approach is to hold the WCGT by the edges and close the wound via pressures outside of the expected WCGT area, and then press the WCGT in place all at once."

In the forgoing description, denigration of the wound-contacting adhesive is a recognized problem since holding at least some of the underside of WCGT 100 is required particularly when side 90 of the wound is drawn toward side

91 of the wound. Thus, from the forgoing description, it appears that wound closure tape only to has a single release liner.

US 2008/0302487 A1 describes a dispensing device configured to operate with an adhesive backed mesh and backing film for tissue bonding. The device prevents or eliminates distortion of the mesh prior to application to the wound sites and includes means for reducing or eliminating binding during use. The dispensing device is configured to operate in a "forward" mode (substrate to which mesh is applied passes beneath applicator after mesh is applied) to provide essentially an unobstructed view of the wound site during use. The backing film is a single strip that protects denigration of adhesive and self adherence of the coiled adhesive-backed mesh.

US 2005/0182443 A is directed to a tissue bonding article which includes a flexible material, an adhesive substance applied over at least a portion of a bottom side of the flexible material, and a polymerizable adhesive composition permeated throughout at least a portion of the flexible material. Although not specifically shown in the figures, a suitable backing or release material may also be used to cover the adhesive substances applied to the bottom side of the flexible material. Such backing materials are well known in the art for covering adhesives and can include, for example, paper, plastic, or the like.

There continues to be a need for improved devices and systems that use surgical tapes, topical adhesives and their combinations such as those provided by this invention.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 depicts a top view of a wound closure device comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 2 depicts a wound-facing view of a wound closure device (prior to removal of the release liners and prior to application to a wound) comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIGS. 3-6 depict edge views of a wound closure device comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 7 depicts a first step for use of the device in accordance with the invention.

FIG. 8 depicts a second step for use of the device in accordance with the invention.

FIG. 9 depicts a third step for use of the device in accordance with the invention.

FIG. 10 depicts a fourth step for use of the device in accordance with the invention.

FIG. 11 depicts a fifth step for use of the device in accordance with the invention.

FIG. 12 depicts an alternate embodiment of the device in accordance with the invention which comprises several perforated segments of individual wound closure devices.

FIG. 13 depicts a bottom view of the wound closure device of FIG. 12 (prior to removal of the release liners and prior to application to a wound) comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 14 depicts a top view of the wound closure device of FIG. 12 comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention

FIGS. 15, 16, 17 and 18 depict edge views of the wound closure device of FIGS. 13 and 14 comprising a non-

3

symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 19 depicts an alternate embodiment of the invention with the bottom view of a wound closure device (prior to removal of the release liners and prior to application to a wound) comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 20 depicts a top view of the wound closure device of FIG. 19 comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention

FIGS. 21-24 depict edge views of the wound closure device depicted in FIGS. 19 and 20 comprising a non-symmetric, two-part release liner in accordance with an embodiment of the invention.

FIG. 25 depicts a first step for use of the device in accordance with the invention.

FIG. 26 depicts a second step for use of the device in accordance with the invention.

FIG. 27 depicts a third step for use of the device in accordance with the invention.

FIG. 28 depicts a fourth step for use of the device in accordance with the invention.

FIG. 29 depicts a fifth step for use of the device in accordance with the invention.

FIG. 30 depicts a sixth step for use of the device in accordance with the invention.

SUMMARY OF THE INVENTION

One aspect of this invention is directed to a wound closure device having a length A, a width B, an upper edge, a lower edge, a right-hand edge, and a left-hand edge comprising:

a wound closure strip having a length A, a width F, a wound-facing side, a top side, an upper edge and a lower edge, the wound-facing side comprising an adhesive applied over at least a portion of the wound facing side;

a release liner assembly detachably adhered to the wound closure strip by the adhesive, the release liner assembly consisting of a first section having a length A and a width E and a second section having a length A and a width D;

wherein the release liner assembly extends the full length A of the wound closure device and extends the full width B of the wound closure device, such that the release liner assembly forms a wound closure strip-free tab, wherein the wound closure strip-free tab has a width of G extending from the lower edge of the wound closure device to the lower edge of the wound closure strip;

wherein the width F of the wound closure strip is greater than the width E of the first section by a width H, where H is equal to length D minus length G; and

wherein the first section and the second section are separated from each other by a liner cut.

In a preferred aspect of the invention,

the wound closure strip is porous;

the length A is from 2 to 20 cm and the width B from 1.7 to 12 cm;

the wound closure strip has a width F extending from the upper edge of the wound closure device and where F is from 1.5 to 8 cm;

the wound closure strip-free tab has a width G of 0.2 to 4.0 cm;

the first section of the release liner assembly has a width of E extending from the upper edge of the wound closure device and is from 1.5 to 8.0 cm and the second

4

section of the release liner assembly has a width D extending from the lower edge of the wound closure device is from 0.2 to 4 cm such that the width E of the first section is greater than the width D of the second section; and

the width B of the device is equal to the width E of the first section plus the width D of the second section.

A further aspect of this invention is directed toward a method of closing a wound having at least a first separated topical tissue surface and a second separated topical tissue surface comprising the steps of:

a) utilizing any one of the embodiments of the inventive wound closure devices described herein and generally comprising a wound closure strip, the wound closure strip comprising a wound facing side and a top side, the wound facing side comprising an adhesive applied over at least a portion of the wound facing side and a non-symmetric, two-part release liner assembly detachably adhered to the adhesive, the release liner assembly comprising a first section and a second section with the second section comprising a wound closure strip-free portion forming a tab;

b) removing the first section of the release liner assembly to expose a portion of the wound facing side of the wound closure strip while grasping the tab of the second section of the release liner assembly;

c) adhering a portion of the exposed wound facing side of the wound closure strip to the at least first separated topical tissue surface and pulling the tab of the second section of the release liner toward the at least second separated topical tissue surface to form two abutted topical tissue surfaces and further adhering the exposed wound facing side of the wound closure strip to the at least second topical tissue surface; and

d) removing the second section of the release liner assembly to further expose a wound facing side of the wound closure strip and adhering the further exposed wound closure strip the second topical tissue surface.

The device used in the method of the invention may be any of the devices disclosed herein.

Desirably the method further comprises the step of applying a flowable, polymerizable adhesive over the adhered wound closure strip.

In alternate embodiments, the invention comprises a wound closure device that is multi-segmented and comprising several individual wound closure devices.

Advantages of this invention include:

1) Minimization of adhesive transfer from the product to the user's fingers/gloves, therefore maximizing grip to tissue.

2) Due to the presence of the second section of the release liner, ease of repositioning the wound closure strip once deployed on approximated wound edges.

3) Adhesive is present on only the side that is intended to be in contact with tissue. So there is no accidental adherence of objects to the wound closure strip side not in contact with tissue.

4) Manipulation of the wound closure strip is stable during tissue approximation as the second release paper section provides sufficient rigidity.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

A key aspect for the improved ease of use of wound closure devices of this invention lies in the non-symmetric, two-part release liner assembly used in conjunction with a wound closure strip.

5

FIGS. 1 and 2 depict a top view and wound-facing view, respectively, of one embodiment of the wound closure device 100 of the present invention having upper edge 1, lower edge 2, right-hand edge 3 and left-hand edge 4. Referring to FIGS. 1 and 2, wound closure device 100 of length A and width B comprises wound closure strip 110 of length A, width F, and lower edge 6; and release liner assembly 140 having a cut 5 and consisting of a first section 120 of length A and width E and second section 130 of length A and width D. It should be noted that dimension C represents the width past the midline of wound closure device 100 that first section 120 extends; dimension H represents the overlap of wound closure strip 110 with second section 130; and dimension G represents the portion 7 of second section 130 that is free of wound closure strip 110 and available for gripping device 100 without having to touch the adhesive containing wound contacting side of wound closure strip 110.

FIGS. 3, 4, 5 and 6 provide an upper edge view, a lower edge view, a left-side edge view and a right-side edge view, respectively, of device 100.

Typical and preferred dimensions for wound closure device 100 are:

Dimension	Typical (cm)	Most Preferred (cm)
A	2.0-20.0	12.0
B	1.0-14	4.5
C	>0.0	1.0
D	0.5-4.0	1.5
E	0.6-10.0	3.0
F	1.0-8.0	3.5
G	0.2-4.0	1.0
H	0.2-2	0.5

With respect to the various dimensions noted above, one skilled in the art would recognize that other dimensions for the wound closure device of this invention are possible depending on the specific dimensions one skilled in the art may select and adapt to treat wounds or incisions of various sizes and shapes. Additionally, for example, referring to FIG. 1, it is envisioned that wound closure strip 110 perimeter p need not be coextensive with portions of device 100 perimeter p', and can be partially coextensive or not be coextensive at all depending if a margin between wound closure strip 110 and release liner assembly 140 is desired.

In general, device 100 may be of any length A and width B that permits the user to effectively apply to close a wound or incision. For many applications, length A will typically range from about 2 cm to about 20 cm, preferably from about 10 cm to about 15 cm, and most preferably about 12 cm. Width B will typically range from about 1.0 cm to about 14 cm, preferably from about 2 to about 10 cm, and most preferably about 4.5 cm.

Additionally, wound closure strip 110 may be of any length A and width F that permits the user to effectively apply to close a wound or incision. For many applications, length A will typically range from about 2 cm to about 20 cm, preferably from about 10 cm to about 15 cm, and most preferably about 12 cm. Width F will typically range from about 1.0 cm to about 8 cm, preferably from about 2 cm to about 6 cm, and most preferably about 3.5 cm.

Dimension C, the distance between the Midline of wound closure device 100 and release liner cut 5 between first section 120 and second section 130. It is dimension C that makes the release assembly a non-symmetric, two-part assembly. If dimension C was equal to 0 cm, then the release

6

assembly would no longer be non-symmetric as then the release assembly cut 5 would be at the Midline and hence the first release section 120 and the second release section 130 would be of equal width. So dimension C should be greater than 0 and ranges from 0.1 cm to about 2 cm, preferably from about 0.5 cm to about 1.5 cm, and most preferably about 1.0 cm.

Dimension D ranges from 0.5 cm to about 4 cm, preferably from about 0.7 cm to about 3 cm, and most preferably about 1.5 cm. It is within the foregoing ranges that device 100 is conveniently handled with sufficient overlap of wound closure strip 110 and second section 130 to adequately accomplish approximation of the separated topical tissue surface of a wound.

Dimension E is the width of the first section 120 and relates to the width of wound closure strip 110 that is suitable for the user to position and/or reposition wound closure strip 110 for the initial approximation of the separated topical tissue surfaces of a wound. Dimension E ranges from 0.6 cm to about 10 cm, preferably from about 2.0 cm to about 5 cm, and most preferably about 3.0 cm. It is within the foregoing ranges that wound closure strip 110 adequately approximates and cover wounds ranging in sizes from 1 cm to about 20 cm.

Dimension G ranges from 0.2 cm to about 4 cm, preferably from about 0.5 cm to about 3 cm, and most preferably about 1.0 cm. It is within the foregoing ranges that device 100 is conveniently handled without denigrating the pressure sensitive adhesive of the wound-face side of wound closure strip 110.

Dimension H ranges from 0.2 cm to about 2.0 cm, preferably from about 0.3 cm to about 1.0 cm, and most preferably about 0.5 cm. It is within the foregoing ranges that device 100 is conveniently handled with sufficient adherence of second section 130 to wound closure strip 110 to accomplish approximation of the wound.

For treatment of surgical wound incisions of about 1.0 cm to about 12 cm, a preferred embodiment of device 100 is for an overall length A of about 12 cm and overall width B of about 4.5 cm, wound closure strip 110 of overall length A of about 12 cm and overall width F of about 3.5 cm, Dimension C of about 1.0 cm, Dimension D of about 1.5 cm, Dimension E of about 3.0 cm, Dimension G of about 1.0 cm and Dimension H of about 0.5 cm.

It would be appreciated by one of skill in the art that device 100 may be cut or trimmed to size for wounds within the above identified range. In practice, it is desirable to have wound closure strip 110 extend approximately at least 0.3 cm, preferably 1 cm from the longitudinal ends of the wound or incision site.

Alternate embodiments of this invention are contemplated wherein the device comprises a wound closure device that is multi-segmented and comprises several individual wound closure devices as shown in FIGS. 12-24. A method of using such devices is shown in FIGS. 25-30.

Referring to FIG. 12, device 100 comprises wound closure strip 110 and release liner assembly 140 having a first section 120 and a second section 130 separated by cut 5. Device 100 further comprises several perforated segments of individual wound closure devices 100a, 100b, 100c, 100d, and 100e. This embodiment demonstrates the versatility of the invention in being able to provide an array of devices that may be more suitable for the closure of smaller incisions, such as those made in laparoscopic procedures having incisions in the range of 0.5-1.5 cm. Also in this embodiment, wound closure strip 110 is coextensive with release liner assembly 140. An alternative embodiment where

wound closure strip **110** is not coextensive with release liner assembly **140** is described below.

FIG. **13** depicts the bottom view of the wound closure device **100** of FIG. **12** (prior to separation from the multi-segmented array, removal of the release liners and application to a wound) of overall segmented length **A'** and width **B** comprising a non-symmetric, two-part release liner. More specifically, segmented devices **110a**, **100b**, **100c**, **100d** and **100e** are separated by perforations **pa**, **pb**, **pc**, and **pd**, respectively. Further referring to segmented device **100a**, this device of length **A** and width **B** comprises top side edge **1**, bottom side edge **2**, left edge **3** and right edge **4**.

FIG. **14** depicts a top view of the wound closure device **100** of FIG. **12** comprising a non-symmetric, two-part release liner assembly **140** comprising first section **120** of width **E** and second section **130** of width **D** separated by cut **5**.

FIGS. **15**, **16**, **17** and **18** depict an upper edge view, a lower edge view, a left-side edge view and a right-side edge view, respectively, of device **100**.

Typical and preferred dimensions for wound closure device **100a** of FIGS. **13** and **14** are:

Dimension	Typical (cm)	Preferred range (cm)	Most Preferred (cm)
A'	5.0-40.0	6.5-25	15
A	1.0-8.0	1.5-5	3
B	1.0-10	1.5-5	3
C	>0.0	0.1-2	1.0
D	0.5-4.0	0.5-3	1.5
E	0.5-6	1-5	1.5

In general, and particularly referring to device **100a** of FIG. **13**, device **100a** may be of any length **A** and width **B** that permits the user to effectively apply to close a wound or incision. Dimension **A'**, the overall length of the multi-segmented device **100**, shall vary as a function of the length **A** varies. For example, for device **100** comprising 5 individual wound closure device segments (i.e., **100a**, **100b**, **100c**, **100d**, and **100e**), length **A'** would be five times the length **A**. For many applications, length **A** will typically range from about 1 cm to about 8 cm, preferably from about 1.5 cm to about 5 cm, and most preferably about 3 cm. Width **B** will typically range from about 1.0 cm to about 12 cm, preferably from about 1.5 to about 7 cm, and most preferably about 3.0 cm.

Additionally, wound closure strip **110a** may be of any length **A** that permits the user to effectively apply to close a wound or incision. For many applications, length **A** will typically range from about 1 cm to about 8 cm, preferably from about 1.5 cm to about 5 cm, and most preferably about 3 cm.

Dimension **C**, is the distance between the Midline of wound closure device **100a** and release liner cut **5** between first section **120** and second section **130**. It is dimension **C** that makes the release assembly a non-symmetric, two-part assembly. If dimension **C** was equal to 0 cm, then the release assembly would no longer be non-symmetric as then the release assembly cut **5** would be at the Midline and hence the first release section **120** and the second release section **130** would be of equal width. So dimension **C** should be greater than 0 and preferably ranges from 0.1 cm to about 2 cm and most preferably about 1.0 cm.

Dimension **D** ranges from 0.5 cm to about 4 cm, preferably from about 0.5 cm to about 3 cm, and most preferably about 1.5 cm. It is within the foregoing ranges that device **100a** is conveniently handled with sufficient overlap of

wound closure strip **110** and second section **130** to adequately accomplish approximation of the separated topical tissue surface of a wound.

Dimension **E** is the width of the first section **120** and relates to the width of wound closure strip **110a** that is suitable for the user to position and/or reposition wound closure strip **110a** for the initial approximation of the separated topical tissue surfaces of a wound. Dimension **E** ranges from 0.5 cm to about 6 cm, preferably from about 1.0 cm to about 5 cm, and most preferably about 1.5 cm. It is within the foregoing ranges that wound closure strip **110a** adequately approximates and cover wounds ranging in sizes from 0.25 cm to about 7 cm.

With respect to the various dimensions noted above, one skilled in the art would recognize that other dimensions for the wound closure device of this invention are possible depending on the specific dimensions one skilled in the art may select and adapt to treat wounds or incisions of various sizes and shapes. Additionally, for example, referring to FIG. **13**, it is envisioned that wound closure strip **110** (**100a**) perimeter need not be coextensive with portions of device **100** (**100a**) perimeter, and can be partially coextensive or not be coextensive at all depending if a margin between wound closure strip **110** and release liner assembly **140** is desired.

For treatment of surgical wound incisions of about 2 cm, a preferred embodiment of device **100a** is for an overall length **A** of about 4 cm and overall width **B** of about 4.5 cm, wound closure strip **110a** of overall length **A** of about 4 cm and overall width **F** of about 3.5 cm, Dimension **C** of about 1.0 cm, Dimension **D** of about 1.5 cm, Dimension **E** of about 3.0 cm, Dimension **G** of about 1.0 cm and Dimension **H** of about 0.5 cm.

It would be appreciated by one of skill in the art that device **100a** may be cut or trimmed to size, or even overlaid for wounds within the above identified ranges. In practice, it is desirable to have wound closure strip **110a** extend approximately at least 0.3 cm, preferably 1 cm from the longitudinal ends of the wound or incision site. Also as an example of an over lay, it is envisioned that a wound of 7 cm in length may be closed using 3 wound closure strips **110a** of 3 cm in length. Over laying multiple wound closure strips **110a** provide an advantage of closing wound that may have curvilinear shapes.

FIG. **19** depicts an alternate embodiment of the invention with the bottom view of a wound closure device **100** (prior to separation from the multi-segmented array, removal of the release liners and application to a wound) comprising a non-symmetric, two-part release liner of overall length **A'** and width. More specifically, segmented devices **110a**, **100b**, **100c**, **100d** and **100e** are separated by perforations **pa**, **pb**, **pc**, and **pd**, respectively. Further referring to segmented device **100a**, this device of length **A** and width **B** comprises top side edge **1**, bottom side edge **2**, left edge **3** and right edge **4**. This embodiment is an example of an instance that wound closure strip **110** perimeter is not coextensive with the perimeter of device **100**.

FIG. **20** depicts a top view of the wound closure device **100** of FIG. **19** comprising a non-symmetric, two-part release liner.

FIGS. **21**, **22**, **23** and **24** depict an upper edge view, a lower edge view, a left-side edge view and a right-side edge view, respectively, of device **100** as depicted in FIGS. **19** and **20**.

Typical and preferred dimensions for wound closure device **100a** of FIGS. **19** and **20** are:

Dimension	Typical (cm)	Preferred range	Most Preferred (cm)
A'	5.0-40.0	6.5-25	15
A	1.0-8.0	1.5-5	3
B	1.0-12	1.5-7	5
C	>0.0	0.1-2	1.0
D	0.5-5	0.5-4	2
E	0.5-7	1-5	3
F	1.0-8.0	1.5-5	3
G	0.25-4.0	0.5-2	1.0
H	0.25-4	0.5-2	1

With respect to the various dimensions noted above, one skilled in the art would recognize that other dimensions for the wound closure device of this invention are possible depending on the specific dimensions one skilled in the art may select and adapt to treat wounds or incisions of various sizes and shapes.

In general, device **100a** may be of any length A and width B that permits the user to effectively apply to close a wound or incision. Dimension A', the overall length of the multi-segmented device **100**, shall vary as a function of the length A varies. For example, for device **100** comprising 5 individual wound closure device segments (i.e., **100a**, **100b**, **100c**, **100d**, and **100e**), length A' would be five times the length A. For many applications, length A will typically range from about 1 cm to about 8 cm, preferably from about 1.5 cm to about 5 cm, and most preferably about 3 cm. Width B will typically range from about 1.0 cm to about 12 cm, preferably from about 1.5 to about 7 cm, and most preferably about 5 cm.

Additionally, wound closure strip **110a** may be of any length A and width F that permits the user to effectively apply to close a wound or incision. For many applications, length A will typically range from about 1 cm to about 8 cm, preferably from about 1.5 cm to about 5 cm, and most preferably about 3 cm. Width F will typically range from about 1.0 cm to about 8 cm, preferably from about 2 cm to about 6 cm, and most preferably about 3.5 cm.

Dimension C, the distance between the Midline of wound closure device **100** and release liner cut **5** between first section **120** and second section **130**. It is dimension C that makes the release assembly a non-symmetric, two-part assembly. If dimension C was equal to 0 cm, then the release assembly would no longer be non-symmetric as then the release assembly cut **5** would be at the Midline and hence the first release section **120** and the second release section **130** would be of equal width. So dimension C should be greater than 0 and ranges from 0.1 cm to about 2 cm, preferably from about 0.5 cm to about 1.5 cm, and most preferably about 1.0 cm.

Dimension D ranges from 0.5 cm to about 5 cm, preferably from about 0.5 cm to about 4 cm, and most preferably about 2 cm. It is within the foregoing ranges that device **100** is conveniently handled with sufficient overlap of wound closure strip **110a** and second section **130** to adequately accomplish approximation of the separated topical tissue surface of a wound.

Dimension E is the width of the first section **120** and relates to the width of wound closure strip **110a** that is suitable for the user to position and/or reposition wound closure strip **110a** for the initial approximation of the separated topical tissue surfaces of a wound. Dimension E ranges from 0.5 cm to about 7 cm, preferably from about 1.0 cm to about 5 cm, and most preferably about 3.0 cm. It is within the foregoing ranges that wound close strip **110a**

adequately approximates and covers wounds ranging in sizes from 0.25 cm to about 7 cm.

Dimension G ranges from 0.25 cm to about 4 cm, preferably from about 0.5 cm to about 2 cm, and most preferably about 1.0 cm. It is within the foregoing ranges that device **100a** is conveniently handled without denigrating the pressure sensitive adhesive of the wound-face side of wound closure strip **110a**.

Dimension H ranges from 0.25 cm to about 4.0 cm, preferably from about 0.5 cm to about 2.0 cm, and most preferably about 1 cm. It is within the foregoing ranges that device **100a** is conveniently handled with sufficient adherence of second section **130** to wound closure strip **110a** to accomplish approximation of the wound.

For treatment of surgical wound incisions of about 2 cm, a preferred embodiment of device **100a** is for an overall length A of about 4 cm and overall width B of about 4.5 cm, wound closure strip **110a** of overall length A of about 4 cm and overall width F of about 3.5 cm, Dimension C of about 1.0 cm, Dimension D of about 1.5 cm, Dimension E of about 3.0 cm, Dimension G of about 1.0 cm and Dimension H of about 0.5 cm.

It would be appreciated by one of skill in the art that device **100a** may be cut or trimmed to size, or even overlaid for wounds within the above identified ranges. In practice, it is desirable to have wound closure strip **110a** extend approximately at least 0.3 cm, preferably 1 cm from the longitudinal ends of the wound or incision site. Also as an example of an over lay, it is envisioned that a wound of 7 cm in length may be closed using 3 wound closure strips **100a** of 3 cm in length. Over laying multiple wound closure strips **110a** provide an advantage of closing wound that may have curvilinear shapes.

Wound closure strips suitable for use in this invention comprise any suitable strip that is adaptable to close a wound. Preferably, the wound closure strip is porous and will allow a flowable, polymerizable adhesive to permeate the strip and to allow adequate bonding of the strip to a tissue surface being bonded.

The wound closure strip comprises a wound facing side and a top side. The wound facing side further comprises an adhesive such as a pressure sensitive adhesive (PSA) applied over at least a portion of the wound facing side. The adhesive may be provided over the entire wound facing side of the wound closure strip. The PSA is useful for initially approximating the wound. The wound closure strip is preferably porous. By "porous" is meant herein either that the bulk of the wound closure strip has pores, such that subsequently applied polymerizable adhesive composition is soaked up or absorbed by the bulk material, or that the bulk of the wound closure strip has voids (like a net or screen), such that the subsequently applied polymerizable adhesive composition passes directly through the bulk material, with or without being soaked up or absorbed by the bulk material. For example, in the case of textile materials, "porous" is generally used to mean that the applied adhesive composition permeates and passes through interstices between the fibers, but does not necessarily pass into and through the fibers themselves. Preferably the wound closure strip is a mesh.

Such porosity (or other properties such as hydrophobicity or hydrophilicity) will also allow a polymerization initiator or rate modifier to be loaded in or on the wound closure strip prior to use, to initiate the subsequently applied polymerizable adhesive composition. Such porosity will also preferably allow air and fluid to pass through the wound closure strip, either through pores per se, or through voids in the

bulk material. Depending upon the degree of porosity and/or the size of the openings, such porosity of the mesh or ability of air and fluid to permeate through the mesh may be tailored either to remain after a final composite material is formed, or to be absent therefrom. The wound closure strip is also preferably non-toxic, as it is intended to be used cover a wound, such as on biological tissues. As such, the wound closure strip should be biologically compatible with the desired substrate (such as tissue, skin, organ, or the like), and is preferably a material that is governmentally approved or generally regarded as safe for the desired purpose. By way of example, suitable wound closure strips are mesh materials and are disclosed in United States Patent Applications 2006/0009099 and 2005/0182443, incorporated herein by reference in their entirety.

Suitable wound closure strip materials may be formed of either synthetic or natural materials. Such material may be formed of either woven or non-woven fabrics or materials. The wound closure strip may be, for example, any suitable polymeric film, plastic foam (including open celled foam), a woven fabric, knitted fabric, a non-woven fabric, mixture thereof, or the like.

In particular, suitable wound closure strips may thus be prepared, for example, from nylon, a polyolefin film, such as polyethylene, polypropylene, ethylene propylene copolymers, and ethylene butylene copolymers, polyurethanes, polyurethane foams, polystyrenes, plasticized polyvinylchlorides, polyesters, polyamides, polylactic acid, polyglycolic acid, polycaprolactone, copolymer mixtures of the above, and cotton. Suitable specific examples include, for example, nylon, polyethylene, polypropylene, ethylene propylene copolymers, ethylene butylene copolymers, polyurethane, polystyrene, plasticized polyvinylchloride, polyester, polyamide, cotton, polytetrafluoroethylene (PTFE), biovascular material, collagen, Gore-Tex®, DACRON®, etc.

The wound closure strip may be formed of a synthetic, semi-synthetic, or natural organic material. Thus, for example, the wound closure strip may be formed of a synthetic or natural polymer material, but not from a material such as metal (such as silver, steel or the like) or glass or ceramic. The wound closure strip may be either biodegradable, or not biodegradable. The wound closure strip is preferably resistant to tearing.

The thickness of the wound closure strip may be from about 0.05 mm to about 10 mm. In another embodiment, the thickness of the wound closure strip is from about 0.1 mm to about 7 mm, preferably from about 0.3 mm to about 5 mm, most preferably from about 0.3 mm to about 3 mm.

The wound closure strip may be selected to be elastic or have some memory effect. In such embodiments, the elastic properties of the mesh may desirably provide a degree of pressure or stress at the application site, for example, to maintain wound edge approximation. Likewise, in embodiments where such additional degree of pressure or stress at the application site is not desired, the mesh may be selected to have less or no elasticity.

The wound closure strip may be either biodegradable, or not biodegradable. By "biodegradable" is meant that the mesh biodegrades over time in vivo, such that it does not require physical removal of the mesh after a set period of time. Thus, for example, a biodegradable mesh is one that, in the in vivo environment, will biodegrade over a period of from about one week to about five years. A non biodegradable material is one that does not biodegrade in an in vivo environment within about five years. Such a non biodegradable material thus would require physical removal of the

wound closure strip at a desired time, rather than slowly deteriorating over time or may slough off naturally from the tissue.

The wound closure strip preferably includes one or more chemical materials located in or on it. For example, one or more chemical substances may be dispersed in or on the wound closure strip, such as being chemically bound, physically bound, absorbed, or adsorbed to it. Thus, for example, the wound closure strip preferably includes at least a polymerization initiator or rate modifier, and may optionally include one or more bioactive materials. As desired, the one or more chemical substances may be either immobilized in or on the wound closure strip, for example, so that it has a desired effect but is not detached from the wound closure strip during use.

For example, a polymerization initiator or rate modifier may be loaded in or on the wound closure strip so that the initiator or rate modifier provides the desired initiation or rate modification effect to a subsequently applied polymerizable adhesive composition. The polymerization initiator or rate modifier may be immobilized in or on the wound closure strip, so that the initiator or rate modifier does not become detached from the wound closure strip and its residues are dispersed in the resultant polymeric material. Alternatively, for example, the polymerization initiator or rate modifier may be initially attached to the wound closure strip, but only in such a manner that it becomes mobilized or solubilized by a subsequently applied polymerizable adhesive composition and dispersed in the resultant polymeric material.

If desired, a combination of chemical substances may also be provided in or on the wound closure strip, to provide multiple effects. For example, as described above, a first chemical species (such as a polymerization initiator or rate modifier) may be immobilized in or on the wound closure strip, while a second, different chemical species (such as a bioactive material) may be detachably attached to the wound closure strip. Other combinations of chemical species and resultant effects are also envisioned.

The chemical substance may be applied in a uniform manner to the wound closure strip, such that there is a substantially uniform concentration of the chemical substance across the wound closure strip. Alternatively, the chemical substance may be applied such that a concentration gradient exists across or through the wound closure strip. For example, a greater or smaller concentration of the chemical substance could exist at the center or edges of the wound closure strip, or a greater or smaller concentration of the chemical substance could be applied on one side of the wound closure strip as compared to an opposite side. Further, the chemical substance may be applied in a uniform manner to the wound closure strip, or it may be applied in a non-uniform random or patterned manner (such as lines, dots, concentric circles, or the like). The chemical substances may also be on, beneath, or in the adhesive layer applied to the wound closure strip.

When present in or on the wound closure strip, the chemical substances (i.e., polymerization initiator, rate modifier, and/or bioactive materials, or other additives), may be incorporated in or on the wound closure strip in any suitable manner. For example, the chemical substance may be wound closure strip added to the wound closure strip by contacting the wound closure strip with a solution, mixture, or the like including the chemical substances. The chemical substance may be added to the wound closure strip, for example, by dipping, spraying, roll coating, gravure coating, brushing, vapor deposition, or the like. Alternatively, the

13

chemical substance may be incorporated into or onto the wound closure strip during manufacture of the wound closure strip, such as during molding, knitting/weaving, scouring, tenting, plaiting or other processing or the like of the wound closure strip.

Other chemical substances that may be present in or on the wound closure strip include, but are not limited to, any suitable and preferably compatible additive that enhances performance of the composite structure. Such additional chemical substances may be bioactive or non-bioactive. Suitable other chemical substances thus include, but are not limited to, colorants (such as inks, dyes and pigments), scents, protective coatings that do not chemically detach, temperature sensitive agents, drugs, wound-healing agents, anti-microbial agents and the like.

The polymerization initiator or rate modifier loaded in or on the wound closure strip may provide a number of advantages for example, the tailoring of the setting or polymerization time of the applied polymerizable adhesive composition. For example, the type and/or concentration of initiator that is applied to the wound closure strip may be selected so as to provide faster or slower polymerization time. The concentration of polymerization initiator or rate modifier may be increased to provide a faster polymerization time, or may be decreased to provide a slower polymerization time.

Because the polymerization initiator or rate modifier is loaded directly in or on the wound closure strip, it is not necessary to mix the polymerizable adhesive composition with a polymerization initiator or rate modifier prior to application. This may allow a longer working time, where the polymerizable monomer composition may be more precisely and carefully applied over a longer period of time.

Such suitable initiators are known in the art and are described, for example, in U.S. Pat. Nos. 5,928,611 and 6,620,846, both incorporated herein by reference in their entireties, and U.S. Patent Application No. 2002/0037310, also incorporated herein by reference in its entirety. Quaternary ammonium chloride and bromide salts useful as polymerization initiators are particularly suitable. By way of example, quaternary ammonium salts such as domiphen bromide, butyrylcholine chloride, benzalkonium bromide, acetyl choline chloride, among others, may be used.

Benzalkonium or benzyltrialkyl ammonium halides such as benzyltrialkyl ammonium chloride may be used. When used, the benzalkonium halide may be benzalkonium halide in its unpurified state, which comprises a mixture of varying chain-length compounds, or it can be any suitable purified compound including those having a chain length of from about 12 to about 18 carbon atoms, including but not limited to C12, C13, C14, C15, C16, C17, and C18 compounds. By way of example, the initiator may be a quaternary ammonium chloride salt such as benzyltrialkyl ammonium chloride (B TAC).

Other initiators or accelerators may also be selected by one of ordinary skill in the art without undue experimentation. Such suitable initiators or accelerators may include, but are not limited to, detergent compositions; surfactants: e.g., nonionic surfactants such as polysorbate 20 (e.g., Tween 20™ from ICI Americas), polysorbate 80 (e.g., Tween 80™ from ICI Americas) and poloxamers, cationic surfactants such as tetrabutylammonium bromide, anionic surfactants such as sodium tetradecyl sulfate, and amphoteric or zwitterionic surfactants such as dodecyltrimethyl(3-sulfopropyl) ammonium hydroxide, inner salt; amines, imines and amides, such as imidazole, arginine and povidine; phosphines, phosphites and phosphonium salts, such as triph-

14

enylphosphine and triethyl phosphite; alcohols such as ethylene glycol, methyl gallate; tannins; inorganic bases and salts, such as sodium bisulfite, calcium sulfate and sodium silicate; sulfur compounds such as thiourea and polysulfides; polymeric cyclic ethers such as monensin, nonactin, crown ethers, calixarenes and polymeric-epoxides; cyclic and acyclic carbonates, such as diethyl carbonate; phase transfer catalysts such as Aliquat 336; organometallics such as cobalt naphthenate and manganese acetylacetonate; and radical initiators or accelerators and radicals, such as di-*t*-butyl peroxide and azobisisobutyronitrile.

Mixtures of two or more, such as three, four, or more, initiators or accelerators may be used. A combination of multiple initiators or accelerators may be beneficial, for example, to tailor the initiator of the polymerizable monomer species. For example, where a blend of monomers is used, a blend of initiators may provide superior results to a single initiator. For example, the blend of initiators can provide one initiator that preferentially initiates one monomer, and a second initiator that preferentially initiates the other monomer, or can provide initiation rates to help ensure that both monomer species are initiated at equivalent, or desired non-equivalent, rates. In this manner, a blend of initiators can help minimize the amount of initiator necessary. Furthermore, a blend of initiators may enhance the polymerization reaction kinetics. The polymerization initiator, accelerator, rate-modifier, and/or cross-linking agent may be incorporated into the mesh using impregnation methods known in the art.

The two-part release liner assembly is a backing film comprising individually peelable sections. The first section **120** is coextensive with at least a portion of wound closure strip **110** and is of larger width compared to the second section **130**. Second section **130** comprises at least a portion that is not coextensive with the wound closure strip **110**. This non-coextensive portion of section **130** permits manipulation and placement of wound closure strip **110** without disturbing or denigrating the pressure sensitive adhesive contained on the wound-facing side of wound closure strip **110**. In some instances, it is contemplated that second section **130** may be wider than first section **120** to aid in manipulation and gripping of the wound closure device as the separated topical tissue surfaces are being approximated.

The material for the invention's release liner assembly may be any suitable backing or release material used to cover the adhesive substances applied to the wound facing side of the wound closure strip. Such backing materials are well known in the art for covering adhesives and can include, for example, paper, plastic, or the like. By way of example, the release liner assembly may be a silicone treated material. Preferably, the release liner assembly is of a material that prevents or eliminates the wound closure strip from sticking to itself.

The method of wound closure or tissue bonding herein disclosed includes a polymerizable adhesive composition applied over the wound closure strip after the wound closure strip is applied to a tissue or wound site. The polymerizable adhesive composition may comprise a polymerizable monomeric adhesive. In embodiments, the polymerizable adhesive composition comprises a polymerizable 1,1-disubstituted ethylene monomer formulation. In embodiments, the polymerizable adhesive composition comprises a cyanoacrylate formulation. In embodiments, synthetic polymerizable adhesive materials such as polyurethane, polyethylene glycol, acrylates, glutaraldehyde and biologically based adhesives may be used.

15

Suitable α -cyanoacrylate monomers which may be used, alone or in combination, include alkyl α -cyanoacrylates such as 2-octyl cyanoacrylate; dodecyl cyanoacrylate; 2-ethylhexyl cyanoacrylate; butyl cyanoacrylate such as n-butyl cyanoacrylate; ethyl cyanoacrylate; methyl cyanoacrylate or other α -cyanoacrylate monomers such as methoxyethyl cyanoacrylate; 2-ethoxyethyl cyanoacrylate; 3-methoxybutyl cyanoacrylate; 2-butoxyethyl cyanoacrylate; 2-isopropoxyethyl cyanoacrylate; and 1-methoxy-2-propyl cyanoacrylate. In embodiments, the monomers are ethyl, n-butyl, or 2-octyl α -cyanoacrylate. Other cyanoacrylate monomers which may be used include alkyl ester cyanoacrylates, such as those prepared by the Knoevenagel reaction of an alkyl cyanoacetate, or an alkyl ester cyanoacetate, with paraformaldehyde, subsequent thermal cracking of the resultant oligomer and distillation.

The wound closure device herein disclosed may be provided in a kit comprising additional components. The kit may comprise at least one wound closure strip as herein described, and one or more containers of polymerizable adhesive composition. The different components or groups of components may be sterilized in separate containers before packaging the components or groups of components within a kit, and thereafter sterilizing the kit as disclosed in co-assigned U.S. Pre-grant Patent Publication No. 2004/0120849, incorporated herein by reference in its entirety.

The adhesive substance used on the wound closure strip may, for example, be any suitable adhesive substance. Preferably, the adhesive substance is a medical grade adhesive, such as acrylic based pressure sensitive adhesives (PSAs), rubber based pressure sensitive adhesives, silicone pressure sensitive adhesives, mixtures thereof, or the like. It is preferred that the adhesive substance be different from the polymerizable adhesive composition. Thus, for example, it is preferred that while the polymerizable adhesive composition can be, for example, a polymerizable monomeric adhesive composition, the adhesive substance is a material that is not a polymerizable adhesive composition, such as a pressure sensitive adhesive.

Suitable rubber based PSAs include, but are not limited to, those taught in U.S. Pat. No. 5,705,551 and in U.S. Pat. No. 4,080,348, the disclosures of which are hereby incorporated by reference. Examples of polymeric rubber bases include one or more of styrene-isoprene-styrene polymers, styrene-olefin-styrene polymers including styrene-ethylene/propylene-styrene polymers, polyisobutylene, styrene-butadiene-styrene polymers, polyisoprene, polybutadiene, natural rubber, silicone rubber, acrylonitrile rubber, nitrile rubber, polyurethane rubber, polyisobutylene rubber, butyl rubber, halobutyl rubber including bromobutyl rubber, butadiene-acrylonitrile rubber, polychloroprene, and styrene-butadiene rubber.

A particularly useful rubber based adhesive is that which has a thermoplastic elastomeric component and a resin component. The thermoplastic elastomeric component contains about 55-85 parts of a simple A-B block copolymer wherein the A-blocks are derived from styrene homologs and the B-blocks are derived from isoprene, and about 15-45 parts of a linear or radical A-B-A block copolymer wherein the A-blocks are derived from styrene or styrene homologs and the B-blocks are derived from conjugated dienes or lower alkenes, the A-blocks in the A-B block copolymer constituting about 10-18 percent by weight of the A-B copolymer and the total A-B and A-B-A copolymers containing about 20 percent or less styrene. The resin component consists of essentially of tackifier resins for the elastomeric component. In general, any compatible

16

conventional tackifier resin or mixture of such resins may be used. These include hydrocarbon resins, rosin and rosin derivatives, polyterpenes and other tackifiers. The adhesive substance may contain about 20-300 parts of the resin component per one hundred parts by weight of the thermoplastic elastomeric component. One such rubber based adhesive substance is commercially available from Ato Findley under the trade name HM3210.

Useful acrylic based PSAs include, but are not limited to, those taught in U.S. Pat. Nos. 5,947,917 and 5,164,444 (acrylic emulsion), U.S. Pat. No. 5,623,011 (tackified acrylic emulsion). It can also be radiation curable mixture of monomers with initiators and other ingredients such as those taught in U.S. Pat. No. 5,232,958 (UV cured acrylic) and U.S. Pat. No. 5,232,958 (EB cured). The disclosures of these patents are hereby incorporated by reference.

It is contemplated that any acrylic based polymer capable of forming an adhesive layer with sufficient tack to adhere to the wound closure strip, the backing film or to a substrate, and with acceptable adhesion to skin, may be used. In certain embodiments, the acrylic polymers for the pressure-sensitive adhesive layers include those formed from polymerization of at least one alkyl acrylate monomer or methacrylate, an unsaturated carboxylic acid and optionally a vinyl lactam. Examples of suitable alkyl acrylate or methacrylate esters include, but are not limited to, butyl acrylate, ethyl acrylate, 2-ethylhexyl acrylate, isooctyl acrylate, isononyl acrylate, isodecyl acrylate, methyl acrylate, methylbutyl acrylate, 4-methyl-2-pentyl acrylate, sec-butyl acrylate, ethyl methacrylate, isodecyl methacrylate, methyl methacrylate, and the like, and mixtures thereof. Examples of suitable ethylenically unsaturated carboxylic acids include, but are not limited to, acrylic acid, methacrylic acid, fumaric acid, itaconic acid, and the like, and mixtures thereof. A preferred ethylenically unsaturated carboxylic acid monomer is acrylic acid. Examples of suitable vinyl lactams include, but are not limited to, N-vinyl caprolactam, 1-vinyl-2-piperidone, 1-vinyl-5-methyl-2-pyrrolidone-, vinyl pyrrolidone, and the like, and mixtures thereof.

Useful silicone pressure sensitive adhesives include those commercially available from Dow Corning Corp., Medical Products and those available from General Electric. Examples of silicone adhesives available from Dow Corning include those sold under the trademarks BIO-PSA X7-3027, BIO-PSA X7-4919, BIO-PSA X7-2685, BIO-PSA X7-3122 and BIO-PSA X7-4502. Additional examples of silicone pressure sensitive adhesives are described in U.S. Pat. Nos. 4,591,622, 4,584,355, 4,585,836 and 4,655,767, the entire disclosures of which are incorporated herein by reference.

The adhesive substance may also include one or more tackifiers, plasticizers, antioxidants, cutting agents such as waxes, and surfactants. Other optional materials that may be added to the adhesive substance layer in minor amounts (typically less than about 25% by weight of the elastomeric phase) include pH controllers, medicaments, bactericides, growth factors, wound healing components such as collagen, antioxidants, deodorants, perfumes, antimicrobials and fungicides.

FIGS. 7 to 11 depict steps for the use of the device of this invention.

In particular and referring to FIG. 7, once a suitable wound W has been identified for closure of tissue T by joining topical tissue surface TS₁ with topical tissue surface TS₂, first section 120 of wound closure device 100 is removed by grasping and pulling section 120 (for example, with left hand, LH) while holding second section 130 (for example with right hand RH) to separate first section 120

17

from wound closure strip **110**. Next, FIG. **8** illustrates the device being positioned axially along a side of wound **W** of tissue **T** and contacting a first separated topical tissue surface TS_1 . FIG. **9** depicts tissue surface TS_1 on one side of wound **W** being pulled to approximate a second separated topical tissue surface TS_2 of wound **W** to the first separated topical tissue surface TS_1 of wound **W** by pulling second section **130**. FIG. **10** shows second section **130** being removed from wound closure strip **110** and wound closure strip **110** being more firmly secured to topical tissue surface TS_2 by the user. Once wound closure strip **110** is firmly contacting tissue **T** and has approximated TS_1 to TS_2 of wound **W**, FIG. **11** illustrates application of a suitable flowable, polymerizable adhesive being applied from applicator **800** which permeates wound closure strip **110** to form film **810**.

FIGS. **25-30** depict typical steps used in a method of applying a segment of the multi-segmented device embodiment to wound **W** of tissue **T** by joining topical tissue surface TS_1 with topical tissue surface TS_2 . Referring to these figures, the user accomplishes the method for the closing of a wound comprising the steps of:

- a) grasping multi-segmented device **100**; (FIG. **25**)
- b) separating segment **100a** from device **100**; (FIG. **26**)
- c) removing first section **120** of release liner **140** from device **100a** to expose wound closure strip **110a** coated with a pressure sensitive adhesive; (FIG. **27**)
- d) positioning wound closure strip **110** axially along a side of wound **W** of tissue **T** and contacting a first separated topical tissue surface TS_1 and pulling wound closure strip **110** across wound **W** to approximate and join topical tissue surface TS_1 with topical tissue surface TS_2 , (FIG. **28**)
- e) removing second section **130** of release liner **140**; (FIG. **29**) and
- f) optionally applying a polymerizable liquid adhesive from applicator **800** to form film **810** over wound closure strip **110**. (FIG. **30**)

Advantages of the methods, devices and systems of this invention include: clear visualization for placement of wound closure strip **110** to approximate wound **W** by abutting the separate topical tissue surfaces of wound **W**; flexibility to reposition wound closure strip **110** and adjust the approximation of wound **W** (i.e., the abutment of the separated topical tissue surfaces of the **W**) before wound closure strip **110** is fully placed on the abutted topical tissue surfaces; ability to handle placement of wound closure strip **110** without denigrating or minimizing denigration of the adhesive side of wound closure strip **110**; manipulation of wound closure strip **110** is more stable during tissue approximation as second section **130** provides rigidity for wound closure strip **110**.

It should be understood that the foregoing disclosure and description of the present invention are illustrative and explanatory thereof and various changes in the size, shape

18

and materials as well as in the description of the preferred embodiment may be made without departing from the spirit of the invention.

What is claimed is:

1. A multi-segmented wound closure device, comprising:
 - a multi-segmented wound closure strip, wherein each segment of the multi-segmented wound closure strip is configured to be separated from the multi-segmented wound closure strip, the multi-segmented wound closure strip having a wound-facing side and a top side opposite the wound-facing side, the wound-facing side comprising an adhesive applied over at least a portion of the wound facing side;
 - a release liner assembly detachably adhered to the wound closure strip by the adhesive, the release liner assembly consisting of a first section and a second section; wherein the release liner assembly extends over an entire length of each segment of the multi-segmented wound closure device and over an entire width of the multi-segmented wound closure device, wherein a width of the wound closure strip is greater than a width of the first section of the release liner assembly; wherein the first section and the second section are separated from each other by a liner cut, and wherein the liner cut is a shared boundary between the first section and the second section.
2. The wound closure device of claim **1** wherein: the wound closure strip is porous.
3. The wound closure device of claim **1**, wherein each segment of the multi-segmented wound closure device has a length of 1-8 cm.
4. The wound closure device according to any one of claim **1**, wherein the wound closure strip is a mesh.
5. The wound closure device according to any one of claim **1**, wherein the adhesive is provided over the entire wound-facing side of the wound closure strip.
6. The wound closure device according to any one of the claim **1**, wherein the adhesive is a medical grade adhesive including one of acrylic based pressure sensitive adhesives (PSAs), rubber based pressure sensitive adhesives, silicone pressure sensitive adhesives and mixtures thereof.
7. The wound closure device of any one of the claim **1** wherein the wound closure strip includes at least a polymerization initiator or rate modifier.
8. A kit comprising:
 - (a) the wound closure device of any one of the claim **1**; and
 - (b) a polymerizable adhesive composition for application over the wound closure strip after the wound closure strip has been applied to a tissue or wound site.
9. The kit according to claim **8**, wherein the polymerizable adhesive composition is different from the adhesive applied to the wound facing side of the wound closure strip.

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